

Recovery Histories: Exact Passage Compression and Additive Choice

Yushang Cheng^{a,*}

^aSouthwest Jiaotong University Hope College, Chengdu, China

ARTICLE INFO

Keywords:

adjustment history
passage compression
additive measurement
directed passage
target attainment
protocol choice

ABSTRACT

This paper studies which features of a one-dimensional target-relative adjustment history survive an additive, chronology-coarse evaluation. A passage account records side, direction, crossed levels, multiplicity, and target attainment. Every weak-time-change-invariant, concatenation-additive valuation factors through a universal passage carrier, and the exact fibers of that carrier are characterized by feasible transformations of actual histories. In one dimension, ordered endpoints together with gross crossing profiles reconstruct the full directed account.

Balanced passage-cycle consistency (BPCC) yields an exact partial additive order on the non-additively-closed path domain. Pairwise vanishing-benchmark consistency (PVBC) characterizes when actual-history comparisons admit an exact unanimity representation by normalized real supporting valuations without imposing a total order on formal differences. Punctured revealed-order closedness (PROC) is a regularity refinement under which the supporting family admits four directed Stieltjes measures while target attainment may retain a boundary atom. Each regular valuation then separates endpoint progress, gross-passage adjustment, and target attainment.

The path geometry also supports controlled economic variation. Given finitely many feasible timed traversals with common endpoints, sufficiently long common horizons allow duration and the cumulative value of any one continuous flow statistic to be matched by adding endpoint stasis alone, without respeeding the traversals or changing their passage accounts. Under free timing, the same matching is available at any prescribed horizon. Equal passage-account differences transfer ordinal rankings across environments, and finite observed protocol choices give sharp bounds for unobserved passage contrasts. A unique common scalarization is a separate codimension-one identification case. Finally, fixed menu cardinality does not bound the multiplicity required for complete additive-consistency audits.

JEL classification: C60; D11; D90; I12.

1. Introduction

Consider Xiaoming choosing among staged smoking-reduction protocols. Two protocols can share their initial state, terminal state, duration, and a cumulative exposure measure while differing in whether they reverse direction, which levels they cross, and whether they reach the treatment target. The paper asks which information in such a history can be compressed without loss for an additive passage evaluation, and what restrictions that compression places on observed choice.

A history first generates a *passage account*. The account records side, direction, crossed levels, multiplicities, and target attainments. It deliberately abstracts from calendar dating and from the order of closed excursions that can be exchanged without changing the directed crossing record. This abstraction is characterized rather than assumed away: every weak-time-change-invariant, concatenation-additive valuation factors through a universal carrier, and Theorem 5.2 proves the converse fiber statement. Two finite-step histories have the same account exactly when feasible time changes, stasis, same-basepoint excursion commutations, and closed-path rotations connect them. The theorem therefore identifies the precise chronology that the account forgets.

The one-dimensional geometry yields a second compression result. Net directional passage is fixed by ordered endpoints, so ordered endpoints together with gross crossing profiles reconstruct the full directed ledger, including the target trace. This identifies a smaller sufficient record and shows that direction is not an independent primitive once endpoint flow and gross passage are jointly observed.

*Corresponding author

✉ 1302573502@qq.com (Y. Cheng)

ORCID(s): 0009-0001-3218-6423 (Y. Cheng)

The account is preference free. Additive measurement begins from actual-history comparisons on a domain that is not closed under arbitrary formal addition. Balanced passage-cycle consistency (BPCC) excludes balanced lists of revealed improvements whose account differences cancel and produces an exact partial additive order. Pairwise vanishing-benchmark consistency (PVBC) excludes actual strict comparisons that are infinitesimal relative to a fixed full-support benchmark. It yields an exact unanimity representation by normalized real supporting valuations while leaving unrevealed formal directions partially ordered.

Regularity is a refinement of that representation. A punctured band $(a, b_n]$ with $b_n \downarrow a$ disappears from the passage record, whereas $[0, b_n]$ continues to contain a target event. Punctured revealed-order closedness (PROC) closes revealed comparisons under the first type of vanishing perturbation. Regular supporting valuations then induce four directed Stieltjes measures, with a possible target atom. Target-oriented directional monotonicity supplies the application-specific orientation of the four modules.

For each regular supporting valuation Λ , one-dimensional crossing conservation gives

$$U_\Lambda(\gamma) = P_\Lambda(\gamma(1)) - P_\Lambda(\gamma(0)) + \sum_s \int V_s^\gamma d\kappa_{s,\Lambda} + \sum_s \alpha_{s,\Lambda} A_s(\gamma).$$

The decomposition is mainly interpretive: endpoint progress is separated from reversal-even gross passage and from target attainment. When several supporting valuations are compatible with the primitive order, these components are valuation specific and observable conclusions are stated in ordinal or set-identified form.

The economic use of the account does not require every background consequence to be literally invariant across arbitrary histories. Proposition 2.1 constructs matched protocol schedules without requiring arbitrarily fast traversal. Given any finite collection of right-side protocols with common endpoints and application-feasible timed traversals, admissible endpoint stasis is enough to match duration and the cumulative value of any one continuous flow statistic at every sufficiently long common horizon, while leaving each traversal and passage account unchanged. When traversal timing is itself free, the horizon restriction disappears. Thus endpoint, duration, and a chosen cumulative exposure can be held fixed by design while passage geometry varies.

Two observable implications follow. First, equal passage-account differences transfer primitive ordinal rankings across protocol environments. Second, on a finite empirical grid, finitely many observed choices define a polytope of additive completions whose linear image gives sharp bounds for any unobserved passage contrast. A monetary numeraire can calibrate the same finite-design bounds in monetary units, but the identification result itself is the interval of compatible passage contrasts rather than a selected scalar utility.

The mathematical tools draw on additive measurement, ordered-vector-space separation, qualitative probability, and Stieltjes representation. The domain-specific results are the exact path fibers, the one-dimensional reconstruction of direction from endpoints and gross passage, additive pricing on a non-additively-closed history domain, the target boundary between vanishing passage and surviving attainment, and path-realizable cancellation certificates whose multiplicity is unbounded even on six histories. Smoking reduction is the running protocol environment; the formal results apply to one-dimensional target-relative adjustment problems with the same passage semantics.

The paper is organized as follows. Sections 2–4 define the economic object and path domain. Section 5 develops the universal account, exact fibers, and one-dimensional compression. Sections 6–7 develop additive supporting valuations and the regular measure refinement. Section 8 gives the endpoint–gross–passage–attainment interpretation and protocol-transfer restriction. Section 9 studies matched designs and finite-data prediction. The appendices collect the fiber proof, separation arguments, regularity construction, and cancellation-complexity result.

2. Target-relative recovery histories

2.1. A target-relative state and four directed movements

Let x_t denote a one-dimensional state measured relative to a predetermined target, with $x_t = 0$ exactly at the target. In the smoking application, one may take $x_t = A_t - A_t^*$, where A_t is an addiction stock and A_t^* is the current treatment target, or use any increasing bounded normalization that preserves the target and the ordering of states. The representation takes x_t as primitive; an application supplies its law of motion.

When $x > 0$, moving toward 0 is reduction from above the target, whereas moving away is relapse escalation. When $x < 0$, moving away is further undershooting or destabilization, whereas moving toward 0 is a return to the planned treatment level. These movements define the four modules T_R , A_R , A_L , and T_L , each of which may carry its own value measure.

2.2. Common accounting and individual-specific valuation

A recovery diary first generates a directed-passage account. The account records the target-relative levels crossed by each module and their multiplicities. The primitive conditional weak order then assigns subjective values to the module-specific profiles, with the non-passage consequences treated as fixed background variables. For one individual, one internally consistent valuation system applies throughout the comparison domain. The same account may receive a different valuation for another individual.

The conditioning defines the comparison held fixed in the passage order. Under a given law of motion, a detour and a direct reduction may carry different consumption, health, or adjustment consequences and therefore need not be jointly feasible while those consequences are literally identical. The conditional order isolates a partial evaluation of the passage account. In a designed or naturally controlled menu where the non-passage consequences are equal within a compared pair, observed choice directly reveals this passage order. When they differ, an application combines the passage order with a model of those consequences. As with procedural and sequence utility (Loewenstein and Prelec, 1993; Frey, Benz and Stutzer, 2004; Frey and Stutzer, 2005), the conditional object can be economically meaningful without requiring passage and consequence to be separately manipulable in every application.

Supporting passage valuations are within-person objects; interpersonal comparison requires an additional normalization.

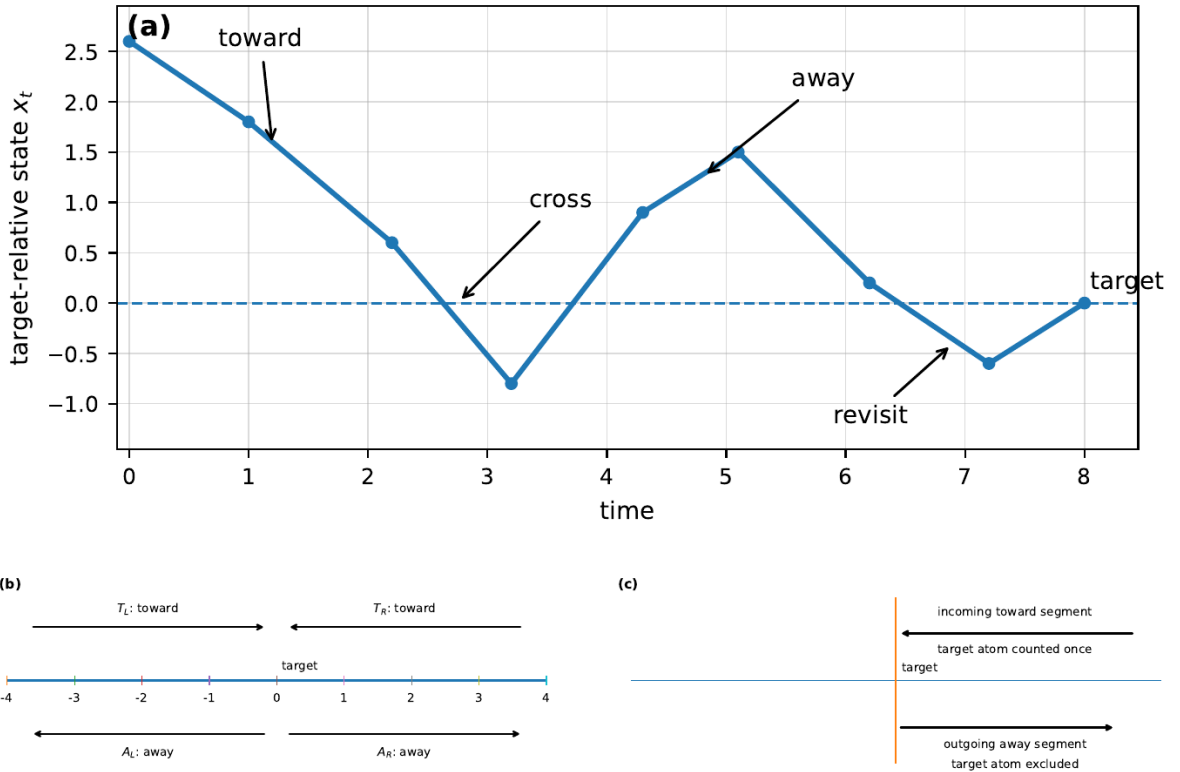


Figure 1: Directed accounting for a target-relative recovery history. Panel (a) shows one ordered path that approaches, crosses, leaves, and revisits the target. Panel (b) assigns nonstationary monotone pieces to the four modules T_L , T_R , A_L , and A_R . Panel (c) gives the target-interface convention. A continuous crossing is one physical movement but two directed passage pieces: the incoming toward piece includes the target and records one target attainment, whereas the outgoing away piece excludes target mass.

2.3. The target interface

Whenever a monotone segment crosses the target, the passage account splits it at 0. The incoming toward segment is target-ending and includes the target point. The outgoing away segment is target-departing and excludes it. Thus a

continuous crossing records exactly one attainment. The two directed pieces may nevertheless carry different punctured values because they lie in different modules.

For example, a continuous movement from $x = 2$ to $x = -2$ is one physical reduction episode. In the account it consists of a right-side toward passage from 2 to 0, including one right-side target attainment, followed by a left-side away passage from 0 to -2 , with target mass excluded. This bookkeeping separates the value of reaching the target from the value of continuing beyond it.

2.4. The three-history experiment

Fix right-side distances $0 < a < b < c$. Consider three histories with common initial distance c and terminal distance b :

$$\gamma^D : c \longrightarrow b, \quad \gamma^C : c \longrightarrow a \longrightarrow b, \quad \gamma^A : c \longrightarrow 0 \longrightarrow b.$$

The direct history γ^D records monotone reduction. The detour history γ^C makes additional progress to a and then relapses to b . The target-touching history γ^A reaches the target and then returns to b . All three have the same ordered endpoints, so their differences isolate internal passage value.

For every normalized regular compatible supporting valuation Λ , the representation below implies

$$\begin{aligned} U_\Lambda(\gamma^C) - U_\Lambda(\gamma^D) &= 2\kappa_{R,\Lambda}((a, b]), \\ U_\Lambda(\gamma^A) - U_\Lambda(\gamma^D) &= 2\kappa_{R,\Lambda}((0, b]) + \alpha_{R,\Lambda}. \end{aligned}$$

At a fixed supporting valuation, the first difference isolates the off-target reduction–relapse detour and the second isolates the extra near-target passage contrast together with target attainment. Across the supporting family these cardinal components are set identified. Sections 8 and 9 turn the same geometry into ordinal transfer restrictions and sharp finite-data bounds. The endpoint-only subclass predicts $\gamma^D \sim \gamma^C \sim \gamma^A$; any strict ranking among them rejects target-relative terminal-state sufficiency.

2.5. Matched duration and flow exposure

The passage record can vary in protocols that are matched on familiar outcome summaries. Let a *geometric protocol* specify the ordered finite-step state path. An application may also supply a feasible timed traversal of that path. The result below keeps each such traversal fixed and uses only stasis at the common endpoints as the matching instrument.

Proposition 2.1 (Matched protocol design with fixed traversals). *Fix $0 \leq b < c$ and finitely many finite-step geometric protocols $\pi^1, \dots, \pi^K$ contained in $[0, c]$, each starting at c and ending at b . For each k , let $z_k : [0, \tau_k] \rightarrow [0, c]$, with $\tau_k > 0$, be a timed traversal of π^k that is feasible in the application. Assume that arbitrary nonnegative stasis at the common endpoints c and b may be appended to each z_k while preserving feasibility. Let $g : [0, c] \rightarrow \mathbb{R}$ be continuous with $g(c) \neq g(b)$. Then there is a finite $\bar{T}$ such that, for every $T > \bar{T}$, there exist feasible schedules $\tilde{\pi}^k : [0, T] \rightarrow [0, c]$ obtained from z_k only by adding endpoint stasis, with*

$$\int_0^T g(\tilde{\pi}^k(t)) \, dt$$

identical for every k . The nonstationary traversal, and hence the passage account, is unchanged for every protocol.

Proof. Put

$$m_k = \int_0^{\tau_k} g(z_k(t)) \, dt, \quad \Delta = g(c) - g(b), \quad a_k = \frac{\tau_k g(b) - m_k}{\Delta}.$$

Choose

$$\bar{T} = 2 \max_{1 \leq k \leq K} \{0, -a_k, \tau_k + a_k\}.$$

For any $T > \bar{T}$, define the initial and terminal dwell times

$$\delta_k = \frac{T}{2} + a_k, \quad \beta_k = \frac{T}{2} - \tau_k - a_k.$$

By the definition of $\bar{T}$, both are strictly positive and $\delta_k + \tau_k + \beta_k = T$. Schedule k stays at c for δ_k , follows the original timed traversal z_k without reparameterization, and then stays at b for β_k . Its cumulative flow is

$$\delta_k g(c) + m_k + \beta_k g(b) = \frac{T}{2} (g(c) + g(b)),$$

where the last equality uses $a_k \Delta = \tau_k g(b) - m_k$. Thus every schedule has the same duration and cumulative flow. Only endpoint stasis has been added, so feasibility is preserved by assumption and the passage account is unchanged. $\square$

Taking $g(x) = x$ matches cumulative target-relative exposure. In particular, any application-feasible timed versions of the direct, off-target detour, and target-touching histories in the preceding subsection can be embedded in sufficiently long schedules with common endpoints, duration, and cumulative exposure without altering their traversal speeds. The condition $g(c) \neq g(b)$ is the nondegeneracy condition that shifting dwell time between the two endpoints changes cumulative flow while leaving duration and passage fixed. If traversal timing is itself freely rescalable, the horizon restriction disappears: for any prescribed $T > 0$, the finitely many traversals may first be assigned a sufficiently short common traversal time and the endpoint-dwell construction then applied. The design matches one chosen continuous flow statistic exactly; additional consequences require additional design variables or an explicit application model.

2.6. Comparison domain versus feasible choice

The representation domain contains the finite-step target-relative histories used to compare passage records and identify their algebraic restrictions. Each treatment problem supplies its own feasible menu, which may be a small subset of this comparison domain. Loop repeatability, reversal opportunities, and target-contact opportunities are therefore application-specific feasibility features.

Controlled comparisons make the passage order empirically interpretable. Proposition 2.1 shows that, once application-feasible traversals are given, duration and one cumulative flow consequence can be matched exactly at sufficiently long common horizons by adding endpoint stasis alone; the traversal speeds need not be changed. Under free timing, the same matching is available at any prescribed horizon. Other consequences can be matched by richer designs, separately measured, or modeled in the application that supplies the feasible menu.

3. Relation to the literature

Smoking cessation, rational addiction, and target-relative recovery. Smoking cessation provides the running protocol environment. Becker and Murphy (1988) model stable forward-looking preferences with an addiction stock and adjacent complementarity, and Becker, Grossman and Murphy (1994) bring that structure to cigarette demand. Subsequent work studies changing myopia (Orphanides and Zervos, 1998), present bias and self-control demand (Gruber and Kőszegi, 2001), withdrawal and quitting decisions (Suranovic, Goldfarb and Leonard, 1999), cue-triggered choice (Bernheim and Rangel, 2004), and backsliding in compulsive consumption (Winston, 1980). The passage order isolates a complementary object: how an agent ranks the directed adjustment record conditional on the other consequences represented in those models.

The predetermined, nonbinding target functions as a treatment reference or staged reduction plateau. Goals can support self-regulation (Hsiaw, 2013; Koch and Nafziger, 2011), and commitment contracts have been studied directly in smoking cessation (Giné, Karlan and Zinman, 2010). The agent may miss, cross, or move away from the target. Endogenous target formation, aspiration revision, and commitment design are application-level extensions.

Reference-dependent and goal-based state utility. Reference-dependent models evaluate outcomes relative to a reference point and allow asymmetric sensitivity across the two sides of that point (Tversky and Kahneman, 1991; Kőszegi and Rabin, 2006). The endpoint-only subclass of the present representation provides the closest point of contact. Within any fixed regular compatible supporting valuation, when $\kappa_L = \kappa_R = 0$ and $\alpha_L = \alpha_R = 0$, passage value reduces to

$$U_\Lambda(\gamma) = P_\Lambda(\gamma(1)) - P_\Lambda(\gamma(0)).$$

With a fixed initial state, each supporting valuation then depends only on the target-centered terminal state. The general representation adds two sources of path value at the supporting-valuation level: the gross-passage adjustment generated by directed crossing multiplicities and a boundary contribution from actual target attainment. The maintained environment takes the target as predetermined, leaving expectations-based reference formation to applications.

Sequences, experiences, and procedural value. Preferences over sequences and extended episodes appear in Loewenstein and Prelec (1993) and Vary and Kahneman (1992). Procedural utility allows the production of an outcome to enter welfare (Frey et al., 2004; Frey and Stutzer, 2005), and Minardi and Savochkin (2024) axiomatize memorable effects of past consumption. The passage component retains side, direction, level, and multiplicity while abstracting from dating, duration, recency, adaptation, and the order of independent closed episodes.

A parametric integral such as $g(c, A) = \int_c^A q(s) ds$ values the crossed levels of one monotone adjustment. Directed passage measures provide an ordinal, nonparametric extension to repeated nonmonotone histories, two target sides, two directions, and target-attainment atoms.

Additive measurement and restricted domains. The route from finite cancellation to additive representation is classical; see Scott (1964), Krantz, Luce, Suppes and Tversky (1971), and Wakker (1989). The finite qualitative-probability obstruction of Kraft, Pratt and Seidenberg (1959) is a close antecedent to using balanced finite certificates rather than pairwise consistency alone. Fishburn (2001) studies the number of distinct comparison pairs required on finite product domains while permitting integer multiplicities within certificates. The arithmetic construction here fixes six ledger columns and lets the primitive coefficient multiplicities diverge across a path-realizable family.

Partial operations are central to extensive measurement. Luce (1967) obtains a finitely additive probability representation when disjoint union is partially defined, and Mackenzie (2021) derives related hypotheses from atom-swarming conditions. The present construction algebraically completes feasible path concatenation through a universal passage group and orders only the revealed cone generated by actual comparisons. Additive representation on restricted domains is studied by Chateauneuf and Wakker (1993), Kopylov (2007), and Qin and Rommeswinkel (2022). Allowing the formal difference order to remain partial connects the real-price step to multi-utility representation rather than to a global Hölder scalarization; see Ok (2002), Dubra, Maccheroni and Ok (2004), and Evren and Ok (2011).

Continuity, atoms, paths, and primitive balances. Monotone-continuity conditions are the classical route from finite to countable additivity in qualitative probability (Villegas, 1964; Chateauneuf and Jaffray, 1984; Chateauneuf, 1985); related continuity restrictions have been used to regularize families of priors (Chateauneuf, Maccheroni, Marinacci and Tallon, 2005). The present continuity condition is path generated: only punctured non-target bands whose geometric intersection is empty are declared negligible, while a target-reaching sequence may retain a singleton residue.

Order-forgetting multiplicity summaries are related to Parikh vectors (Parikh, 1966), while path signatures retain noncommutative order information (Chen, 1954; Hambly and Lyons, 2010). The fiber theorem specializes Euler-trail transformations (Abrham and Kotzig, 1980; Fleischner, Sabidussi and Wenger, 1992) to labelled one-dimensional paths with fixed endpoint imbalance and target-attainment labels. Gross level-crossing counts have a classical analytic history (Łochowski, 2017); here they are inputs to preference representation and the endpoint–gross-passage decomposition.

Circuits, conformational indecomposability, and Graver bases are standard (Graver, 1975). Related complexity results include Berstein and Onn (2009) and Tatakis and Thoma (2015). The present primitive element is itself a circuit, has six columns for every parameter value, and grows through arithmetic coefficients of reachable path accounts. The gross-passage adjustment is a static reversal-even component. Entry–exit irreversibility (Dixit, 1989), dynamic hysteresis (Mielke and Roubíček, 2015), and potential–cycle decompositions in games and rankings (Candogan, Menache, Ozdaglar and Parrilo, 2011; Jiang, Lim, Yao and Ye, 2011) provide neighboring but distinct structures.

4. Recovery paths and directed ledgers

Although the formal results apply to any one-dimensional target-relative adjustment, the running interpretation is an addiction stock relative to a staged treatment target. The formal path is a recovery diary; reachability asks whether its proposed threshold crossings and endpoints can belong to one continuous history.

4.1. Paths and modules

Let

$$X = [\underline{x}, \bar{x}] \subset \mathbb{R}, \quad \underline{x} < 0 < \bar{x},$$

with fixed target 0. Write $R_L = -\underline{x}$ and $R_R = \bar{x}$. A *finite-step path* is a continuous map $\gamma : [0, 1] \rightarrow X$ for which there is a finite partition of $[0, 1]$ such that γ is monotone on every cell. Constant pieces are allowed.

The four passage modules are

$$\mathcal{M} = \{T_L, T_R, A_L, A_R\}.$$

The symbol T denotes movement toward the target and A movement away from it. The subscript denotes the side. For $m \in \mathcal{M}$, let R_m be the maximal target distance on the relevant side.

For $0 \leq a < b \leq R_m$, let $p_m(a, b)$ be the canonical monotone passage through the distance interval from a to b . A toward passage moves from distance b to a ; an away passage moves from a to b . The endpoint convention is asymmetric:

- a toward passage $p_{T_s}(0, b)$ records the target attainment and therefore covers $[0, b]$;
- an away passage $p_{A_s}(0, b)$ records departure through $(0, b]$ and carries no separate point passage at 0.

4.2. Passage profiles

Each monotone piece contributes a step function in its module. Summing over all pieces gives the directed passage profiles

$$h_\gamma^m : [0, R_m] \rightarrow \mathbb{N}_0, \quad m \in \mathcal{M}.$$

For a punctured interval $(a, b]$, $h_\gamma^m(r)$ counts how many module- m pieces cross level r . At the target, $h_\gamma^{T_s}(0)$ is the number of attainments from side s , while $h_\gamma^{A_s}(0) = 0$. The concrete ledger is

$$\ell(\gamma) = (h_\gamma^{T_L}, h_\gamma^{T_R}, h_\gamma^{A_L}, h_\gamma^{A_R}) \in \text{Led}.$$

Formally, Led is the product of four integer step-function groups. A toward-side group is generated by $\mathbf{1}_{[0,b]}$ and $\mathbf{1}_{(a,b]}$ with $0 < a < b$, while an away-side group is generated by $\mathbf{1}_{(a,b]}$ with $0 \leq a < b$. Actual ledgers form the nonnegative reachable subset of this group. The order of individual pieces and the time spent traversing them are not recorded. Multiplicity is retained.

A finite ledger induces a finite labelled directed multigraph on the states appearing at its breakpoints. Each unit crossing of an adjacent state interval is an arc labelled by side and direction.

Proposition 4.1 (Feasibility of a recorded passage account). *Let L be a finite four-module step ledger and let $s, t \in X$ be prescribed ordered endpoints.*

1. *If $L \neq 0$, then L is generated by a single finite-step path from s to t if and only if its nonisolated labelled multigraph is connected,*

$$\deg^+(v) - \deg^-(v) = \mathbf{1}_{\{v=s\}} - \mathbf{1}_{\{v=t\}}$$

for every vertex v , and, when $s = t$, the common endpoint s belongs to the nonisolated support.

2. *If $L = 0$, then L is generated by a path from s to t if and only if $s = t$, in which case the realizing paths are constant up to weak time change.*

For a nonzero ledger satisfying the conditions in part 1, an Euler trail of the labelled multigraph gives a realizing path.

Proof sketch. A realizing path traverses every labelled arc with its recorded multiplicity, forcing connected nonisolated support and the displayed endpoint imbalance. Conversely, after replacing multiplicities by parallel arcs, these conditions are exactly the directed Euler-trail criterion. Reading an Euler trail and interpolating adjacent states gives a realizing finite-step path; the zero ledger is realized only by a constant path. Appendix 12 gives the full proof. $\square$

Proposition 4.1 is the basic domain restriction. Formal sums of ledgers need not satisfy its connectivity or endpoint-balance conditions, so the actual path domain is not additively closed.

4.3. Preferences

The primitive preference $\succeq$ is the conditional weak order on the set $\mathcal{P}$ of all finite-step paths, with non-passage consequences held fixed. The constant path at x is denoted c_x . Unless a result explicitly restricts attention to a finite menu, $\mathcal{P}$ denotes the full finite-step domain on X . In particular, it contains all constant paths, reversals, endpoint-compatible concatenations, canonical passages, canonical off-target round trips, target-touching loops, and subdivisions of every canonical interval into any prescribed finite number of nonempty canonical intervals. The assumptions separate into four roles. The weak order and BPCC govern additive consistency of actual comparisons. PVBC and PROC govern real completion and punctured continuity. Canonical interval richness is a domain condition. Directional monotonicity supplies the economic orientation of the target-relative modules.

Assumption 4.2 (Target-oriented directional monotonicity). For every nonempty canonical interval I , each toward passage $p_{T_s}(I)$ is strictly preferred to the constant target path c_0 , while c_0 is strictly preferred to each away passage $p_{A_s}(I)$. Under BPCC all constant paths have the same zero passage account and are therefore indifferent.

Directional monotonicity makes the predetermined target behaviorally oriented: conditional on the maintained background consequences, local movement toward it is preferred to stasis and local movement away is worse. In the regular representation it implies nonnegative target-attainment residues, while strict positivity of an attainment residue remains an additional feature of the preference. BPCC derives indifference on equal-ledger fibers.

5. The universal passage account and passage-equivalent diaries

The first task is to separate the common description of a passage from its individual valuation. Specifying where movement occurs, on which side, in which direction, and with what multiplicity, a path still leaves its utility undetermined. The universal theorem identifies the additive information space through which every admitted valuation must pass; the fiber theorem then characterizes exactly when two ordered diaries deliver the same input to every valuation in that class.

5.1. The universal additive carrier

For every $m \in \mathcal{M}$ and $0 \leq a < b \leq R_m$, introduce a formal generator $e_m(a, b)$. Let $\mathbb{Z}^{(\mathcal{E})}$ be the free abelian group on these generators and quotient by the subdivision relations

$$e_m(a, c) = e_m(a, b) + e_m(b, c), \quad 0 \leq a < b < c \leq R_m.$$

The resulting group is denoted $\mathfrak{L}$. Cutting a path at its turning points and target crossings defines

$$q : \mathcal{P} \rightarrow \mathfrak{L}$$

by summing the corresponding generator classes. Subdivision makes q independent of the chosen cut.

A weak time change is composition with a continuous, nondecreasing, surjective map $[0, 1] \rightarrow [0, 1]$. It may change speed and insert stasis but cannot reverse the order of visited states.

For an abelian group G , let $\text{Val}_G(\mathcal{P})$ denote the set of maps $\Phi : \mathcal{P} \rightarrow G$ that are invariant under weak time changes and additive under every endpoint-compatible concatenation.

Theorem 5.1 (Universal passage account for additive valuations). *Let G be any abelian group. A map $\Phi : \mathcal{P} \rightarrow G$ is invariant under weak time changes and additive under endpoint-compatible concatenation if and only if there is a unique homomorphism $\bar{\Phi} : \mathfrak{L} \rightarrow G$ such that*

$$\Phi = \bar{\Phi} \circ q.$$

Equivalently,

$$\text{Hom}_{\mathbf{Ab}}(\mathfrak{L}, G) \cong \text{Val}_G(\mathcal{P}).$$

Moreover, $\mathfrak{L}$ is canonically isomorphic to the four-module integer step-ledger group, and under this isomorphism q becomes ℓ .

Being valuation-free, the theorem accommodates both individual- and module-specific weights. For individual i , any admitted passage valuation has the form

$$\Phi_i(\gamma) = \bar{\Phi}_i(q(\gamma)),$$

with an individual-specific homomorphism $\bar{\Phi}_i$. What is common is the account $q(\gamma)$; what differs is its valuation.

The zero passage value of constants follows from $c_x * c_x = c_x$ and additivity. Level-, duration-, health-, consumption-, and other non-passage consequences remain fixed in the conditional passage ranking.

Theorem 5.1 uses the full domain $\mathcal{P}$, which contains every canonical generator and every concatenation used by subdivision. Restricted menus require their own richness conditions.

5.2. Exact fibers of the passage account

Let $\approx_p$ be the equivalence relation generated by:

1. weak time changes, including the insertion or deletion of stasis;
2. identification of constant paths as the zero-passage class;
3. commutation of two closed excursions based at the same state;
4. cyclic rotation of a closed path, corresponding to a different recording origin for the same closed diary.

Each operation preserves the directed account. The converse is subtler, requiring equal ledger coefficients to be connected through operations whose intermediate objects remain feasible single paths and whose attainment labels survive unchanged.

Theorem 5.2 (Passage-equivalent recovery diaries). *For finite-step paths on the fixed-target interval,*

$$q(\gamma) = q(\eta) \iff \gamma \approx_p \eta.$$

Proceeding in three geometric steps, the proof first reduces finitely many continuous paths to vertex words on a common labelled path graph, then uses flow imbalance to recover the ordered endpoints of every open realization, and finally applies rooted leaf removal to write each realization as a reduced core path carrying a multiset of closed leaf excursions. Excursion commutation and insertion-point invariance align those multisets. Cyclic rotation is needed only to choose a common base point for a nonzero closed ledger. If the removed leaf is the target, the restored excursion contains exactly one target-ending toward cell and one target-departing away cell, so the attainment convention is preserved. Appendix 13 gives the full argument. Figure 2 illustrates the same-basepoint excursion-commutation generator.

Every additive passage valuation assigns the same passage value to ledger-equivalent histories. The operations above exhaust the information discarded by the directed-passage account; non-passage consequences may still distinguish the histories in a larger application model.

The result uses finite one-dimensional path geometry. Classical segment-reversal and detachment-absorption results permit broader transformations (Abrham and Kotzig, 1980; Fleischer et al., 1992); the present theorem instead gives an exact fiber description using account-preserving operations that stay inside the actual path domain.

5.3. One-dimensional compression from gross passage and endpoints

For side $s \in \{L, R\}$ and punctured distance $r > 0$, let $N_s^T(r; \gamma)$ and $N_s^A(r; \gamma)$ be the toward and away crossing multiplicities. Define

$$D_s^\gamma = N_s^T - N_s^A, \quad V_s^\gamma = N_s^T + N_s^A.$$

Let $\chi_s(x, r)$ equal one when x lies on side s at target distance at least r , and zero otherwise.

Lemma 5.3 (Net-passage conservation). *For every finite-step path and every punctured distance level,*

$$D_s^\gamma(r) = \chi_s(\gamma(0), r) - \chi_s(\gamma(1), r).$$

Thus net directional passage is determined by ordered endpoints.

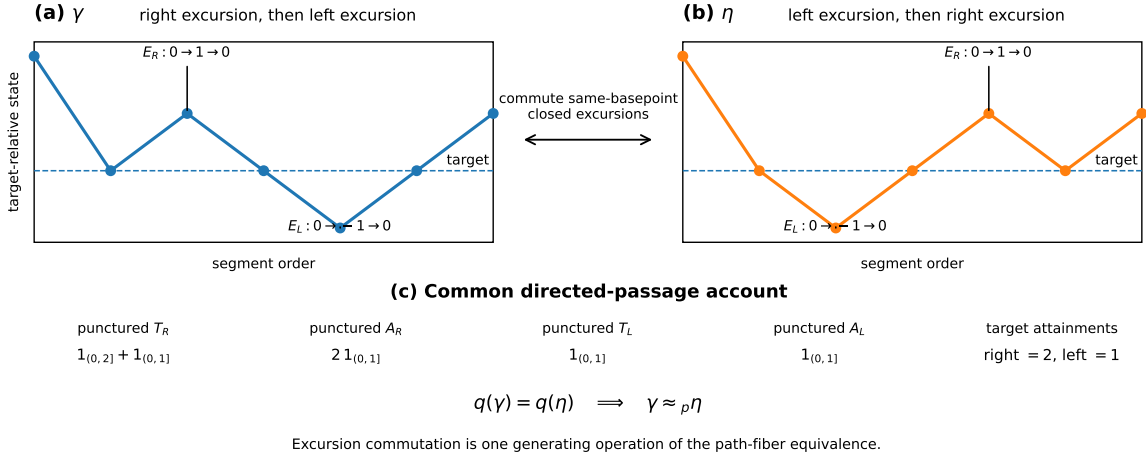


Figure 2: Excursion commutation within a path fiber. The two ordered diaries share the same open backbone and contain the same two closed excursions based at the target, but in opposite orders. Commuting these same-basepoint excursions preserves every punctured directed profile and each side-specific target-attainment count. Hence $q(\gamma) = q(\eta)$ and $\gamma \approx_p \eta$. The figure illustrates one generator of the path-fiber equivalence.

Proposition 5.4 (Lossless reconstruction from gross passage and ordered endpoints). *On the finite-step one-dimensional path domain, the record*

$$(V_L^\gamma, V_R^\gamma, \gamma(0), \gamma(1))$$

reconstructs the full directed passage account $q(\gamma)$. Hence, under BPCC, equality of this record implies primitive indifference.

Proof. Lemma 5.3 recovers D_s from the ordered endpoints, and therefore

$$N_s^T = (V_s + D_s)/2, \quad N_s^A = (V_s - D_s)/2.$$

For a finite-step path, N_s^T is constant on a sufficiently small punctured neighborhood of zero. Its stable near-target value equals the number of toward passages that actually reach the target from side s . The punctured directed profiles and target traces are therefore recovered, which is exactly the full ledger. $\square$

Gross passage alone is not sufficient under strict directional monotonicity: a canonical toward passage and its reverse-away passage through the same punctured band have identical gross profiles but lie on opposite sides of stasis. Ordered endpoints supply the missing directional imbalance. This reconstruction is specific to one-dimensional finite-step geometry.

6. Global consistency and supporting passage valuations

The universal account describes passages without valuing them. The next question is whether one individual's comparisons of actual histories support coherent additive valuations on the account space. BPCC makes the revealed additive order exact on actual histories while preserving partiality on unrevealed formal differences. PVBC characterizes when Archimedean completion preserves those actual-history comparisons. The resulting representation is generally a family of normalized real supporting valuations.

6.1. The canonical ordered envelope

Let $\succeq$ be a weak order on $\mathcal{P}$. Define the comparison cone

$$C_{\succeq} = \left\{ \sum_{i=1}^n (q(\gamma_i) - q(\eta_i)) : \gamma_i \succeq \eta_i \right\} \subseteq \mathfrak{L},$$

the symmetric subgroup

$$\mathcal{N}_{\succeq} = C_{\succeq} \cap (-C_{\succeq}),$$

and the quotient

$$G_{\succeq} = \mathfrak{L} / \mathcal{N}_{\succeq}.$$

The image of the comparison cone defines a partial order on $G_{\succeq}$. Write $J_{\succeq} = \pi_{\succeq} \circ q$.

The finite sums in $C_{\succeq}$ are consistency certificates, aggregating account differences revealed by comparisons of actual histories even when the resulting formal sum lies outside the feasible-history domain.

Definition 6.1 (Balanced passage-cycle consistency (BPCC)). BPCC holds if there are no finite lists $\gamma_i \succeq \eta_i$, with at least one strict comparison, such that

$$\sum_i q(\gamma_i) = \sum_i q(\eta_i).$$

Proposition 6.2 (Canonical ordered envelope and BPCC exactness). *For every weak order, $G_{\succeq}$ is a partially ordered abelian group and $J_{\succeq}$ is monotone, weak-time-change invariant, and additive under concatenation. BPCC holds if and only if*

$$\gamma \succeq \eta \iff J_{\succeq}(\gamma) \geq J_{\succeq}(\eta).$$

Every monotone admissible valuation into an ordered abelian group factors uniquely through $J_{\succeq}$ by a positive homomorphism.

The proposition gives BPCC three precise roles. It prevents balanced strict-improvement cycles, makes the canonical map exact on actual histories, and makes the canonical quotient universal among monotone additive valuations. In particular, the one-comparison case gives

$$q(\gamma) = q(\eta) \implies \gamma \sim \eta.$$

Thus preference indifference on a passage fiber is derived from BPCC rather than imposed as a primitive axiom.

6.2. Pairwise real separation on the actual-history order

For the real separation step, write

$$d(\gamma, \eta) = q(\gamma) - q(\eta), \quad E = \mathfrak{L} \otimes_{\mathbb{Z}} \mathbb{R},$$

and define the real revealed cone and indifference space

$$C = \text{cone}\{d(\gamma, \eta) : \gamma \succeq \eta\}, \quad I = \text{span}_{\mathbb{R}}\{d(\gamma, \eta) : \gamma \sim \eta\}.$$

Every finite collection of passage vectors lies on a common finite grid, so real conic certificates can be rationalized and cleared of denominators.

Proposition 6.3 (Exact realified revealed cone). *Under BPCC,*

$$C \cap (-C) = I,$$

and, for actual histories,

$$d(\gamma, \eta) \in C \iff \gamma \succeq \eta.$$

Pass to $V = E/I$ and $K = C/I$. For a nonempty canonical interval J , define its favorable account difference by

$$x_m(J) = \begin{cases} d(p_m(J), c_0), & m \in \{T_L, T_R\}, \\ d(c_0, p_m(J)), & m \in \{A_L, A_R\}. \end{cases}$$

Under Assumption 4.2, choose one full-range J_m in every module and set

$$u = \sum_{m \in \mathcal{M}} x_m(J_m), \quad e = u + I \in V.$$

Subdivision makes e an order unit: every $v \in V$ is bounded between $-re$ and re for some finite $r > 0$. Any two full-support benchmarks generate the same completion because they dominate one another up to finite positive multiples.

Define the Archimedean closure

$$K^a = \{v \in V : v + n^{-1}e \in K \text{ for every } n \in \mathbb{N}\},$$

its lineality space $N = K^a \cap (-K^a)$, and the Archimedeanized order-unit space

$$\hat{V} = V/N, \quad \hat{K} = K^a/N, \quad \hat{e} = e + N.$$

Let $\rho : V \rightarrow \hat{V}$ be the quotient map and write

$$\hat{d}_{\gamma\eta} = \rho(d(\gamma, \eta) + I).$$

Definition 6.4 (Pairwise vanishing-benchmark consistency (PVBC)). PVBC holds if, for every pair of actual histories γ, η ,

$$d(\gamma, \eta) + I \in K^a \implies \gamma \geq \eta.$$

Equivalently, if $\eta > \gamma$, there is a finite repetition number $n = n(\eta, \gamma)$ for which one fixed full-support benchmark no longer supports the derived comparison that n copies of γ compensate n copies of η . No common repetition bound is required across strict comparisons.

For Xiaoming, PVBC says that a genuine disadvantage between two smoking-cessation histories eventually outweighs one fixed recovery benchmark when that same disadvantage is repeated often enough. The repetition count is pair specific and may be arbitrarily large. This rules out infinitesimal strict rankings on actual histories while leaving unrevealed formal portfolios partially ordered and closed-excision values unrestricted.

A normalized supporting valuation is a linear functional $\varphi : \hat{V} \rightarrow \mathbb{R}$ satisfying

$$\varphi(\hat{K}) \subseteq \mathbb{R}_+, \quad \varphi(\hat{e}) = 1.$$

Let S be the set of all such valuations and define

$$U_\varphi(\gamma) = \varphi(\rho(q(\gamma) + I)).$$

Theorem 6.5 (Pairwise real separation on actual histories). Assume BPCC and strict directional monotonicity. PVBC holds if and only if the normalized supporting family S exactly represents the actual path order:

$$\gamma \geq \eta \iff U_\varphi(\gamma) \geq U_\varphi(\eta) \text{ for every } \varphi \in S.$$

The family S is nonempty, convex, and compact in the topology of pointwise convergence on $\hat{V}$. If $\eta > \gamma$, every normalized supporting valuation weakly ranks η above γ and at least one ranks it strictly above.

Proof sketch. The cone $\hat{K}$ is closed, pointed, and has the order unit $\hat{e}$. Positive separation therefore gives

$$\hat{v} \in \hat{K} \iff \varphi(\hat{v}) \geq 0 \quad \forall \varphi \in S.$$

Actual weak comparisons lie in $K \subseteq K^a$. Conversely, unanimity places an actual difference in K^a , and PVBC returns the primitive weak comparison. The reverse argument shows necessity. Appendix 14 gives the real-cone, Archimedeanization, separation, and compactness details. $\square$

Different normalized supporting valuations may disagree on formal directions that are not fixed by actual-history comparisons, while their unanimity recovers the primitive order on actual histories. An individual U_φ can therefore tie a primitive strict comparison; strictness is guaranteed only somewhere in the supporting family. The singleton case below is the special case in which one normalized valuation is order faithful by itself.

Remark 6.6 (Finite grids do not make PVBC automatic). A finite physical grid makes the account space finite dimensional but leaves path multiplicities unbounded. Hence it does not by itself remove infinitesimal strict comparisons. For example, on a grid with one favorable toward cell T and its unfavorable reverse-away cell A , assign values in the lexicographically ordered group $\mathbb{Z}^2$ by

$$\Phi(T) = (1, 0), \quad \Phi(A) = (-1, 1).$$

Rank actual histories by the lexicographic value of their passage accounts, extending any remaining favorable cells by vectors with positive first coordinate. Then $T > 0 > A$ and BPCC holds because the preference is induced by an additive linearly ordered abelian group. The closed loop $T + A$ has value $(0, 1) >_{\text{lex}} 0$, yet no finite repetition of that loop outweighs any full-support benchmark with positive first coordinate. Thus BPCC and directional monotonicity can hold on a finite grid while PVBC fails. A finite empirical design is different: finitely many observed inequalities generate a polyhedral completion problem even though the unrestricted finite-grid path order need not be Archimedean.

BPCC is the finite-list consistency condition underlying passage additivity. Appendix 15 measures its global depth by constructing six-alternative families whose primitive balanced certificates require unbounded multiplicity. This finite-certificate logic is close to qualitative-probability cancellation (Kraft et al., 1959; Chateauneuf and Jaffray, 1984), while the separating-family representation connects to multi-utility theory (Ok, 2002; Dubra et al., 2004; Evren and Ok, 2011).

7. Punctured revealed-order continuity and target attainment

The real supporting family leaves open a continuity question at the target boundary. A canonical passage through a punctured band $(a, b_n]$ with $b_n \downarrow a$ disappears from the passage record. A target-reaching passage through $[0, b_n]$ continues to contain the attainment event. PROC requires revealed comparisons to be closed with respect to the first type of vanishing passage. Appendix 16 constructs the associated locally convex topology and identifies its continuous positive functionals.

Assumption 7.1 (Canonical interval richness). For every module m , every nonempty canonical interval I , and every integer $q \geq 1$, the interval I can be partitioned into q nonempty canonical intervals whose passages belong to $\mathcal{P}$. The side ranges $[0, R_m]$ are finite.

For each module m , let $\hat{x}_m(J) = \rho(x_m(J) + I) \in \hat{K}$ denote the favorable canonical passage through a punctured interval J . Call

$$\sigma = (m, (I_n)_{n \geq 1})$$

an admissible punctured shrinking sequence when $I_n = (a, b_n] \downarrow \emptyset$ with $b_n \downarrow a$, where $a \geq 0$ for away modules and $a > 0$ for toward modules. Target-reaching sequences $[0, b_n] \downarrow \{0\}$ form a separate boundary class.

Appendix 16 turns this geometric notion of disappearance into a locally convex topology τ° . The punctured non-target bands above converge to zero; target-reaching sequences remain in the boundary class. A normalized supporting valuation $\varphi \in \mathcal{S}$ is *punctured regular* if

$$\varphi(\hat{x}_m(I_n)) \longrightarrow 0$$

for every admissible punctured shrinking sequence. Let $\mathcal{S}_{\text{reg}}$ denote the regular supporting valuations. Appendix 16 proves that these are exactly the continuous normalized positive functionals for τ° .

Assumption 7.2 (Punctured revealed-order closedness (PROC)). For every pair of actual histories γ, η ,

$$\hat{d}_{\gamma\eta} \in \overline{\hat{K}}^{\tau^\circ} \implies \gamma \succeq \eta.$$

PROC has a pairwise behavioral interpretation: if revealed weak comparisons approach an actual pair through non-target passage fragments that vanish from the record, the limiting actual pair preserves the weak ranking. For Xiaoming, bands such as $(10, 10 + \varepsilon_n]$ with $\varepsilon_n \downarrow 0$ vanish as adjustment events, whereas reaching the treatment target remains an event at the boundary.

Theorem 7.3 (Regular directed-measure representation with target attainment). *Assume BPCC, strict directional monotonicity, PVBC, and Assumption 7.1. Then PROC holds if and only if the regular supporting family exactly represents the actual path order:*

$$\gamma \succeq \eta \iff U_\varphi(\gamma) \geq U_\varphi(\eta) \text{ for every } \varphi \in S_{\text{reg}}.$$

When these conditions hold, every $\varphi \in S_{\text{reg}}$ induces a unique normalized four-measure system

$$\Lambda_\varphi = (\lambda_{T_L}^\varphi, \lambda_{T_R}^\varphi, \lambda_{A_L}^\varphi, \lambda_{A_R}^\varphi)$$

of finite nonnegative Borel measures on the corresponding side ranges such that

$$\begin{aligned} U_\varphi(\gamma) = & \int h_\gamma^{T_L} d\lambda_{T_L}^\varphi + \int h_\gamma^{T_R} d\lambda_{T_R}^\varphi \\ & - \int h_\gamma^{A_L} d\lambda_{A_L}^\varphi - \int h_\gamma^{A_R} d\lambda_{A_R}^\varphi, \end{aligned}$$

with

$$\lambda_{A_s}^\varphi(\{0\}) = 0, \quad \lambda_{T_s}^\varphi = \alpha_s^\varphi \delta_0 + \lambda_{T_s}^{\varphi, \circ}, \quad \alpha_s^\varphi \geq 0.$$

Let $\mathfrak{M}_{\text{reg}}(\succeq)$ be the family of these normalized measure systems. Then

$$\gamma \succeq \eta \iff U_\Lambda(\gamma) \geq U_\Lambda(\eta) \text{ for every } \Lambda \in \mathfrak{M}_{\text{reg}}(\succeq).$$

The family may contain more than one normalized supporting valuation and need not be compact. Strict directional monotonicity makes every canonical favorable passage nonnegative throughout the regular family and strictly positive under at least one member.

Proof sketch. Let $K^\circ = \overline{\widehat{K}}^{\tau^\circ}$. If an actual difference lies outside K° , locally convex separation supplies a continuous positive functional that is strictly negative on that difference. Normalizing at the order unit gives a regular state because the continuous dual of τ° is exactly the punctured-regular dual. PROC therefore makes regular states sufficient for exact separation, and the converse follows by continuity. For a fixed regular supporting valuation, subdivision gives finitely additive interval contents and punctured regularity gives right-continuity from above. The ordinary Lebesgue–Stieltjes construction extends the away contents directly. On a toward side, the monotone boundary residue

$$\alpha_s^\varphi = \lim_{b \downarrow 0} U_\varphi(p_{T_s}(0, b)) \geq 0$$

is assigned to the target singleton. Because target-reaching sequences form the boundary class rather than the punctured vanishing class, the atom may survive. Appendix 16 gives the separation and measure-extension details. $\square$

For a fixed regular supporting valuation, $\lambda_m((a, b])$ is the value assigned to crossing the band $(a, b]$ in module m . Different regular supporting valuations can place different finite weights on the same passage while remaining jointly consistent with the primitive order. Densities and parametric kernels, when they exist, are derived restrictions on a particular compatible supporting valuation.

The target-attainment atom is the boundary value attached to reaching the prescribed target from side s under a given regular supporting valuation. The full value of a target-touching episode also includes its punctured toward and away passages. For example, under supporting valuation Λ , one continuous crossing from a right-side distance b to a left-side distance a has passage value

$$\lambda_{T_R}^\Lambda([0, b]) - \lambda_{A_L}^\Lambda((0, a]) = \alpha_R^\Lambda + \lambda_{T_R}^{\Lambda, \circ}((0, b]) - \lambda_{A_L}^\Lambda((0, a]).$$

Thus one physical crossing contributes two directed passage components and one right-side attainment event.

Define the regular-price null space

$$J_{\text{reg}} = \{z \in E : \varphi(\rho(z + I)) = 0 \text{ for every } \varphi \in S_{\text{reg}}\}.$$

Corollary 7.4 (Point identification of one common scalarization). *Under Theorem 7.3, the following are equivalent:*

1. *the normalized regular supporting family $\mathcal{S}_{\text{reg}}$ is a singleton;*
2. *the normalized Stieltjes family $\mathfrak{M}_{\text{reg}}(\geq)$ is a singleton;*
3. *$\text{codim } J_{\text{reg}} = 1$ in E .*

When these conditions hold, unnormalized regular compatible supporting valuations form one positive ray, so the four directed measures and attainment atoms are jointly identified up to one common positive scalar before normalization.

On a fixed finite physical grid, admissible empty-shrinking interval sequences are eventually empty, so the punctured-continuity part of PROC becomes vacuous. PVBC need not become redundant because finite-dimensional account spaces still admit unbounded path multiplicities and non-Archimedean orders; Remark 6.6 gives an explicit example. Point identification on the unrestricted path domain therefore remains the codimension condition above. For a finite empirical design, uniqueness of the normalized additive completion is instead a finite-dimensional polyhedral singleton question determined by the active observed restrictions.

Monotone continuity is the classical route from finite to countable additivity in qualitative probability (Villegas, 1964; Chateaufneuf and Jaffray, 1984). Here the continuity class is generated by punctured passage disappearance. The continuous positive supporting valuations suffice to separate actual histories, while the target singleton remains available for a boundary atom.

8. Endpoint progress, gross-passage adjustment, and target attainment

This section interprets each regular supporting valuation by separating the part fixed by ordered endpoints from a reversal-even adjustment attached to gross traversal and from the boundary contribution of target attainment. Under Theorem 7.3, fix $\Lambda \in \mathfrak{M}_{\text{reg}}(\geq)$ and suppress its superscript within the algebraic derivation. The decomposition applies valuation by valuation; ordinal protocol results below are stated for the primitive order or for sets of compatible completions.

All turns, returns, and repetitions enter through the gross profiles V_s^γ defined in Section 5.3; Lemma 5.3 identifies the corresponding net profiles from ordered endpoints.

Writing

$$\lambda_{T_s}^\circ = \lambda_{T_s}|_{(0, R_s]}, \quad \lambda_{A_s}^\circ = \lambda_{A_s}|_{(0, R_s]},$$

for the punctured restrictions, define

$$\mu_s = (\lambda_{T_s}^\circ + \lambda_{A_s}^\circ)/2, \quad \kappa_s = (\lambda_{T_s}^\circ - \lambda_{A_s}^\circ)/2.$$

Then $\kappa_s \ll \mu_s$ and

$$\theta_s = d\kappa_s / d\mu_s$$

exists with $|\theta_s| \leq 1$ μ_s -almost everywhere. Normalize $P(0) = 0$ and define, for x on side s ,

$$P(x) = -\mu_s((0, |x|]).$$

Let $A_s(\gamma) = h_\gamma^{T_s}(0)$ be the target-attainment count from side s .

Theorem 8.1 (Potential–gross-passage–attainment decomposition). *For every normalized regular compatible supporting valuation Λ , normalize $P_\Lambda(0) = 0$. Its passage value has the decomposition*

$$U_\Lambda(\gamma) = P_\Lambda(\gamma(1)) - P_\Lambda(\gamma(0)) + \sum_{s \in \{L, R\}} \int V_s^\gamma d\kappa_{s, \Lambda} + \sum_{s \in \{L, R\}} \alpha_{s, \Lambda} A_s(\gamma).$$

The three terms are endpoint progress, a gross-passage adjustment, and a target-attainment component. Equivalently, the middle term is

$$\sum_s \int V_s^\gamma \theta_{s, \Lambda} d\mu_{s, \Lambda}.$$

For each fixed Λ , the transformation between its normalized four directed measures and $(P_\Lambda, \kappa_{L,\Lambda}, \kappa_{R,\Lambda}, \alpha_{L,\Lambda}, \alpha_{R,\Lambda})$ is invertible. Under Corollary 7.4, the normalized decomposition is point identified; otherwise the theorem describes the full compatible family valuation by valuation.

Proof. On each side,

$$\int N_s^T d\lambda_{T_s}^\circ - \int N_s^A d\lambda_{A_s}^\circ = \int D_s^\gamma d\mu_s + \int V_s^\gamma d\kappa_s.$$

Lemma 5.3 turns the first integral into $P(\gamma(1)) - P(\gamma(0))$. The target atoms contribute $\alpha_s A_s(\gamma)$. For the fixed supporting valuation, invertibility of the definitions of μ_s , κ_s , and α_s , together with $P(0) = 0$, gives the stated one-to-one reparameterization. $\square$

The distinction in names carries substantive content. The signed measure κ_s is the *gross-passage adjustment measure*: it records the common reversal-even premium or penalty that remains after the endpoint component has been removed. Because V_s^γ is generally nonzero on an open monotone path, the middle functional is a general gross-passage adjustment. For a closed off-target loop through a band E , the endpoint contribution is zero and the remaining value $2\kappa_s(E)$ is the *cycle-passage value*. The phrase “passage hysteresis” refers here to this static reversal-even adjustment; dynamic hysteresis is a separate structure.

Corollary 8.2 (Direct, detour, and target-touching histories). *Fix $0 < a < b < c \leq R_R$ and the right-side histories*

$$\gamma^D : c \rightarrow b, \quad \gamma^C : c \rightarrow a \rightarrow b, \quad \gamma^A : c \rightarrow 0 \rightarrow b.$$

Their full decompositions are

$$\begin{aligned} U(\gamma^D) &= P(b) - P(c) + \kappa_R((b, c]), \\ U(\gamma^C) &= P(b) - P(c) + \kappa_R((b, c]) + 2\kappa_R((a, b]), \\ U(\gamma^A) &= P(b) - P(c) + \kappa_R((b, c]) + 2\kappa_R((0, b]) + \alpha_R. \end{aligned}$$

Consequently,

$$\begin{aligned} U(\gamma^C) - U(\gamma^D) &= 2\kappa_R((a, b]), \\ U(\gamma^A) - U(\gamma^D) &= 2\kappa_R((0, b]) + \alpha_R. \end{aligned}$$

The first difference is the cycle-passage value of the additional off-target reduction–relapse detour. The second is the gross-passage adjustment of the additional target-touching detour plus one attainment from the right. The analogous formulas hold on the left.

Proof. The gross profiles are

$$\begin{aligned} V_R^{\gamma^D} &= \mathbf{1}_{(b,c]}, \\ V_R^{\gamma^C} &= \mathbf{1}_{(b,c]} + 2\mathbf{1}_{(a,b]}, \\ V_R^{\gamma^A} &= \mathbf{1}_{(b,c]} + 2\mathbf{1}_{(0,b]}. \end{aligned}$$

All three histories have ordered endpoints (c, b) , so their endpoint contribution is the common term $P(b) - P(c)$. Only γ^A reaches the target, once from the right. Substitution into Theorem 8.1 gives the three full formulas and hence the two differences. $\square$

The reparameterization is algebraically simple; its interpretation comes from the path-geometric separation of endpoint, repeated passage, and attainment components.

8.1. Counterfactual protocol transfer without selecting a valuation

The decomposition gives a cardinal description of each supporting valuation, but additive passage accounting also yields a purely ordinal restriction that is more useful for controlled protocol choice. Two actual pairs that generate the same account difference must have the same primitive ranking. This statement uses neither point identification nor a selected member of the supporting family.

Theorem 8.3 (Actual-pair difference invariance and protocol transfer). *Assume BPCC. If four actual histories satisfy*

$$q(\gamma) - q(\eta) = q(\gamma') - q(\eta'),$$

then

$$\gamma \geq \eta \iff \gamma' \geq \eta'.$$

The same equivalence holds for strict preference and indifference.

In particular, fix $a > 0$ and two right-side protocol environments $j = 1, 2$ with $0 < a < b_j < c_j$. Define

$$\gamma_j^A : c_j \rightarrow 0 \rightarrow b_j, \quad \gamma_j^C : c_j \rightarrow a \rightarrow b_j.$$

Then

$$q(\gamma_1^A) - q(\gamma_1^C) = q(\gamma_2^A) - q(\gamma_2^C),$$

and therefore the target-touching protocol has exactly the same weak/strict/indifferent ranking relative to the a -turning protocol in both environments. If non-passage consequences are controlled within each pair, one observed protocol ranking predicts the other even though the initial and terminal states differ across environments.

Proof. By Proposition 6.3, under BPCC an actual difference belongs to the revealed cone C exactly when the corresponding weak comparison holds. Equal account differences therefore imply equivalence of the two weak comparisons. Applying the same argument to the reversed differences gives equivalence of the reverse weak comparisons, hence also strict preference and indifference.

For the displayed target-contact protocols, the common portions cancel. Relative to γ_j^C , the path γ_j^A adds one toward target-reaching passage over $[0, a]$ and replaces the away passage over $(a, b_j]$ by an away passage over $(0, b_j]$. The resulting account difference is the same target-reaching/away contrast on $(0, a]$ for both j , independent of b_j and c_j . $\square$

At every regular supporting valuation the common difference in Theorem 8.3 has value

$$2\kappa_{R,\Lambda}((0, a]) + \alpha_{R,\Lambda}.$$

This identity interprets the transferred comparison across the supporting family. Endpoint-only passage preference sets the common contrast to zero and predicts indifference in both menus. A strict controlled choice in one menu therefore rejects that subclass and, under BPCC, predicts the corresponding strict choice in every other actual menu realizing the same passage-account difference.

Corollary 8.4 (Off-target loops and shrinking target-return loops). *For a canonical off-target round trip through $(a, b]$ on side s ,*

$$U(C_{s;a,b}) = 2\kappa_s((a, b]).$$

For a target-return loop of radius ϵ that departs from the target into side s and returns from that side,

$$U(C_{\epsilon,s}) = 2\kappa_s((0, \epsilon]) + \alpha_s \longrightarrow \alpha_s \quad \text{as } \epsilon \downarrow 0.$$

Hence, at any fixed supporting valuation, off-target loops isolate the gross-passage adjustment measure and shrinking target-return loops isolate the attainment residue after the punctured adjustment vanishes. Finite ordinal data identify these quantities through the compatible family rather than through one valuation-level identity alone.

Corollary 8.5 (Endpoint-only subclass). *Under Theorem 7.3 on the full path domain, the primitive passage preference depends only on ordered endpoints if and only if, for every $\Lambda \in \mathfrak{M}_{\text{reg}}(\geq)$,*

$$\kappa_{L,\Lambda} = \kappa_{R,\Lambda} = 0, \quad \alpha_{L,\Lambda} = \alpha_{R,\Lambda} = 0.$$

The two sides may still have different potential measures, and different regular supporting valuations may imply different target-centered endpoint potentials unless Corollary 7.4 holds.

Proof. If the primitive passage preference depends only on ordered endpoints, every off-target round trip and every target-return loop is indifferent to stasis. Exact unanimity then makes each supporting valuation zero on those loops. Corollary 8.4 gives $\kappa_s = 0$ on all punctured intervals and then $\alpha_s = 0$. Conversely, if every supporting valuation has $\kappa_L = \kappa_R = 0$ and $\alpha_L = \alpha_R = 0$, Theorem 8.1 makes every supporting value depend only on ordered endpoints; unanimity in Theorem 7.3 gives the same property for the primitive order. $\square$

For a fixed initial state, this subclass is a target-centered terminal-state order. Theorem 8.1 isolates the additional history content carried by gross directed passage and actual target attainment.

9. Matched protocols and finite-data prediction

The representation results describe the full conditional passage order. Economic applications typically observe only a finite collection of protocol choices. This section works directly with those observations. The main result gives sharp counterfactual passage bounds from ordinal protocol data. A short money-calibration corollary records how the same finite-design geometry can be expressed in monetary units.

9.1. Sharp bounds from ordinal protocol choices

Fix a finite physical grid Π . Let $x_1, \dots, x_J$ be the favorable one-cell passage differences on that grid: toward cells are oriented passage-versus-stasis and away cells are oriented stasis-versus-passage. They span the finite-dimensional grid account space E_Π . Put

$$e_\Pi = \sum_{j=1}^J x_j.$$

Let $\mathcal{O} = \{(\gamma_i, \eta_i)\}_{i=1}^n$ be observed weak protocol choices on the grid, with the maintained background attributes controlled within each observed pair, and write $d_i = q(\gamma_i) - q(\eta_i)$. Define

$$\mathcal{P}_\Pi(\mathcal{O}) = \{p \in E_\Pi^* : p(e_\Pi) = 1, \quad p(x_j) \geq 0 \ \forall j, \quad p(d_i) \geq 0 \ \forall i\}.$$

The weak cell inequalities give the closed target-oriented completion of the finite design.

For an unobserved protocol pair (γ, η) on the same grid, let $d = q(\gamma) - q(\eta)$ and define

$$\underline{\Delta}_\mathcal{O}(d) = \min_{p \in \mathcal{P}_\Pi(\mathcal{O})} p(d), \quad \overline{\Delta}_\mathcal{O}(d) = \max_{p \in \mathcal{P}_\Pi(\mathcal{O})} p(d).$$

Theorem 9.1 (Sharp finite-design counterfactual passage bounds). *Suppose $\mathcal{P}_\Pi(\mathcal{O}) \neq \emptyset$. Then it is a compact convex polytope and, for every counterfactual difference d ,*

$$\{p(d) : p \in \mathcal{P}_\Pi(\mathcal{O})\} = [\underline{\Delta}_\mathcal{O}(d), \overline{\Delta}_\mathcal{O}(d)].$$

The endpoints solve two linear programs. Hence a positive lower bound robustly ranks γ above η , a negative upper bound robustly ranks η above γ , and an interval containing both signs records counterfactual ambiguity across compatible additive completions.

Proof. Because $p(x_j) \geq 0$ and $p(e_\Pi) = \sum_j p(x_j) = 1$, every cell value lies in $[0, 1]$. The x_j span E_Π , so the normalized functionals are bounded; the remaining restrictions are finitely many closed linear inequalities. Thus $\mathcal{P}_\Pi(\mathcal{O})$ is a compact convex polytope. Its linear image under $p \mapsto p(d)$ is a compact interval, with endpoints attained by the two stated linear programs. Every element of the polytope defines a weakly target-oriented additive completion of the finite observations, which establishes sharpness. $\square$

Theorem 8.3 is the exact-transfer case: when an observed pair and a counterfactual pair have the same passage-account difference, the same additive restriction applies to both. The finite-data theorem also combines information from several detour, reversal, and target-contact choices when no single observed comparison has the desired difference.

9.2. Money calibration

Finite protocol-price data can calibrate the same additive contrasts in monetary units without selecting one supporting valuation for the full primitive order. Let an observed object be (γ, m) and maintain quasilinear money within this finite design. The compatible additive passage valuations satisfy

$$p(q(\gamma_i) - q(\eta_i)) \geq m_i - n_i$$

for every observed weak choice $(\gamma_i, m_i) \geq (\eta_i, n_i)$, together with target-orientation inequalities on the chosen grid.

Corollary 9.2 (Sharp money-calibrated passage bounds). *For any counterfactual passage difference d , the set of money-calibrated values $\{p(d)\}$ over the compatible finite-design polyhedron is an interval, possibly unbounded, whose finite endpoints solve two linear programs. If γ carries an additional fee c relative to η , a lower bound strictly above c gives robust strict preference for γ , while an upper bound strictly below c gives robust strict preference for η .*

The corollary is a calibration of the finite-design identified set. The paper's economic comparison does not depend on a global quasilinear completion, and the protocol-price calculation is not used in the full-domain representation theorem.

9.3. Identification and scope

A fixed finite physical grid eliminates nontrivial punctured shrinking sequences, so the PROC regularity issue becomes vacuous on that grid. PVBC remains substantive when the unrestricted path domain allows unbounded repetition: Remark 6.6 gives a lexicographic finite-grid order satisfying finite cancellation and target orientation but violating PVBC. By contrast, a finite empirical design uses only finitely many observed inequalities, and its additive completion problem is polyhedral by construction.

With ordinal protocol data, the decision-relevant identified object is generally

$$[\underline{\Delta}_O(d), \overline{\Delta}_O(d)].$$

With protocol-price data, the same finite-design inequalities calibrate the contrast interval in monetary units. Both ordinal and money-calibrated intervals are decision relevant without separately identifying α_s , κ_s , or P . Exact indifferences can collapse the ordinal compatible polytope. On a d -dimensional design, $d - 1$ independent indifference equalities together with normalization are sufficient to pin down a unique feasible additive completion; more generally, singleton identification is checked from the full active constraint system.

The conditional passage order is most directly observed in controlled protocol comparisons. Ordinal transfer and the normalized finite-grid bounds apply when maintained background attributes are matched or separately netted out. Protocol-price menus provide one explicit way to relax equal monetary cost: the fee enters through the quasilinear numeraire while the passage component remains set identified. Other health, duration, and dynamic consequences can be added through application-specific structure when their values are known or modeled.

BPCC remains a global finite-list restriction for the unrestricted path domain. Appendix 15 gives a six-history family whose primitive balanced certificates have unbounded multiplicity. In finite empirical designs, additive feasibility is checked directly through the corresponding system of linear inequalities.

10. Conclusion

The paper develops a passage-based compression of one-dimensional target-relative histories and an additive measurement theory on the resulting account space. The universal carrier identifies the additive object, while the exact fiber theorem characterizes the feasible chronology transformations that leave it unchanged. Ordered endpoints and gross crossing profiles then reconstruct the full directed ledger in the finite-step one-dimensional domain.

BPCC turns actual-history comparisons into an exact partial additive order. PVBC characterizes when Archimedean completion preserves the actual-history order and therefore admits an exact unanimity representation by normalized real supporting valuations. The finite-grid lexicographic example shows that this restriction remains substantive even when the physical state grid is finite if path repetition is unbounded. PROC supplies a boundary-sensitive regularity refinement, producing directed Stieltjes measures with a possible target atom.

The measure representation yields an endpoint–gross–passage–attainment interpretation valuation by valuation. The economic content is stated at the level of primitive rankings and identified sets. For any finite family of application-feasible traversals with common endpoints, sufficiently long matched schedules can hold duration and one continuous

cumulative flow statistic fixed by adding endpoint stasis alone, without respeeding the traversals; free timing removes the horizon bound. Equal account differences transfer ordinal rankings across environments, and finite observed choices give sharp bounds for counterfactual passage contrasts; protocol-price data can express the same bounds in monetary units.

The account intentionally treats exchangeable closed excursions as chronology neutral. The fiber theorem makes that restriction exact: it identifies the information lost by the compression rather than treating all histories with the same endpoint as equivalent. Order-dependent valuation of excursions, first-attainment effects, endogenous targets, and dynamic state transitions require richer records and provide natural extensions beyond the present additive passage domain.

11. Terminology, notation, and scope conventions

This appendix is a reader's map rather than an inventory of every proof-local symbol. Table 1 records the logical spine of the representation; Table 2 gathers the terms used across more than one main-text section. Symbols confined to a single proof are defined where they arise.

Table 2: Core terminology, notation, and scope conventions

Category	Term or symbol	Meaning and maintained convention
<i>Economic object and path domain</i>		
Conditional passage preference	$\succeq; \{U_\Lambda\}_{\Lambda \in \mathfrak{M}_{\text{reg}}}$	Individual subjective ranking of the target-relative passage account, conditional on non-passage consequences being held fixed; the ranking is recovered by unanimity across regular supporting valuations.
Supporting valuation	U_Λ	One normalized additive functional consistent with the primitive order. Unless the regular family is a singleton, an individual U_Λ need not preserve every primitive strict comparison and is not a standalone utility representation.
Controlled protocol menu	—	A menu in which non-passage consequences are equal within a compared pair, so observed full choice reveals the conditional passage order.
Comparison path domain	$\mathcal{P}$	Rich finite-step domain used for representation; a particular empirical protocol menu may be a small subset.
Target-relative state	x_t , target 0	One-dimensional state measured relative to a predetermined target; in the smoking application, $x_t = A_t - A_t^*$ or an order-preserving normalization.
Side and direction	$s \in \{L, R\}; T, A$	L/R denote the below-/above-target sides; toward reduces target distance and away increases it. These are geometric labels.
Target attainment	$A_s(\gamma)$	Arrival at 0 by an incoming toward passage from side s . The maintained atom is paid per attainment.
Target interface	—	The incoming toward cell includes 0, whereas the outgoing away cell excludes it; a continuous crossing gives two directed pieces and one attainment.
Finite-step path	$\gamma : [0, 1] \rightarrow X$	Continuous path that is monotone on finitely many cells; constant cells are allowed.
Weak time change	—	Continuous nondecreasing surjection of $[0, 1]$ preserving the order of visited states while allowing speed changes and stasis.
<i>Passage account and canonical order</i>		
Abstract passage account	$q(\gamma) \in \mathfrak{Q}$	Universal additive carrier obtained from module-interval generators modulo subdivision.

Continued on next page

Table 2 continued

Category	Term or symbol	Meaning and maintained convention
Concrete ledger	$\mathcal{L}(\gamma) = (h_\gamma^m)_{m \in \mathcal{M}}$	Step-function representation recording side, direction, crossed levels, multiplicity, and target attainments. Theorem 5.1 identifies it with $q(\gamma)$.
Passage-equivalent	$\gamma \approx_p \eta$	Histories connected by the account-preserving operations in Theorem 5.2; equivalently, $q(\gamma) = q(\eta)$.
BPCC	Balanced passage-cycle consistency	Excludes a finite list of weak improvements, at least one strict, whose aggregate passage accounts balance exactly.
Canonical envelope	$G_\succeq, J_\succeq$	Partially ordered quotient generated by revealed account differences and its canonical path map.
PVBC	Pairwise vanishing-benchmark consistency	Requires Archimedean completion to preserve actual-history comparisons; it excludes actual infinitesimal strict differences without totalizing formal differences.
Normalized supporting valuations	$\mathcal{S}$	Positive linear functionals on the Archimedeanized order-unit space, normalized at one full-support benchmark. Their unanimity represents the actual order under Theorem 6.5.
<i>Continuity and directed measures</i>		
Punctured topology	τ°	Finest locally convex topology in which canonical non-target passages shrinking to the empty set converge to zero; target-reaching sequences are excluded.
PROC	Punctured revealed-order closedness	Requires punctured topological closure of the revealed cone not to change actual-history comparisons.
Regular supporting valuations	$\mathcal{S}_{\text{reg}}$	τ° -continuous normalized supporting valuations; equivalently, valuations whose values vanish on every admissible punctured shrinking sequence.
Directed measures	$\lambda_{T_s}^\Lambda, \lambda_{A_s}^\Lambda$	Finite nonnegative Borel measures induced by a regular supporting valuation Λ ; away target mass is normalized to zero.
Punctured measures	$\lambda_{T_s}^{\Lambda, \circ}, \lambda_{A_s}^{\Lambda, \circ}$	Restrictions of a regular supporting valuation's directed measures to positive target distances $(0, R_s]$.
Attainment coefficient and atom	$\alpha_s^\Lambda; \alpha_s^\Lambda \delta_0$	Supporting-valuation-specific boundary residue and its singleton-measure form; it may be zero and need not be common across valuations.
<i>Finite-data protocol choice</i>		
Protocol-price completion	$v_p(\gamma, m) = p(q(\gamma)) - m$	Application-specific quasilinear extension used to calibrate passage values in monetary units on a finite grid.
Willingness-to-pay interval	$[\underline{w}(d), \overline{w}(d)]$	Sharp set of monetary values assigned to a counterfactual protocol contrast across additive target-oriented completions consistent with observed protocol-price choices.
<i>Decomposition and identification</i>		
Net passage	$D_s^\gamma = N_s^T - N_s^A$	Directional imbalance determined by ordered endpoints.
Gross passage	$V_s^\gamma = N_s^T + N_s^A$	Toward-plus-away multiplicity entering the reversal-even adjustment and the information results.
Potential measure and potential	$\mu_s; P$	$\mu_s = (\lambda_{T_s}^\circ + \lambda_{A_s}^\circ)/2$ generates endpoint progress, with $P(0) = 0$.
Gross-passage adjustment	$\kappa_s; \int V_s^\gamma d\kappa_s$	Signed reversal-even component, where $\kappa_s = (\lambda_{T_s}^\circ - \lambda_{A_s}^\circ)/2$.
Cycle-passage value	$2\kappa_s(E)$	Gross-passage adjustment of a closed off-target round trip through E after the endpoint term cancels.
Target-return loop	$C_{\varepsilon, s}$	Departure from the target to radius ε and return from side s ; its value converges to α_s .

Continued on next page

Table 2 continued

Category	Term or symbol	Meaning and maintained convention
Endpoint-only subclass	—	Equivalent to $\kappa_L = \kappa_R = 0$ and $\alpha_L = \alpha_R = 0$; left and right potential measures may still differ.

Conditions or inherited result	Role in the representation chain
Finite-step path domain; weak-time-change invariance; concatenation additivity	Universal carrier through which every admitted group-valued passage valuation factors (Theorem 5.1).
One-dimensional labelled path geometry	Exact feasible fibers of the carrier (Theorem 5.2).
Ordered endpoints + gross passage in the finite-step one-dimensional domain	Lossless reconstruction of the full directed ledger (Proposition 5.4).
Weak order + BPCC	Exact canonical partial additive order on actual histories (Proposition 6.2).
+ Target-oriented directional monotonicity + PVBC	Exact unanimity representation by normalized real supporting valuations, without totality of formal differences (Theorem 6.5).
+ Interval richness + PROC	Regular supporting valuations suffice for the actual order; each induces four directed Stieltjes measures and side-specific target atoms (Theorem 7.3).
+ $\text{codim } J_{\text{reg}} = 1$	Point identification of one normalized common scalarization (Corollary 7.4).
Theorem 7.3 + net-passage conservation	Endpoint–gross-passage–attainment decomposition of each supporting valuation (Theorem 8.1).
Common endpoints + fixed feasible traversals + admissible endpoint stasis + one continuous flow statistic	For sufficiently long common horizons, matched duration and flow without respeeding; free timing removes the horizon bound (Proposition 2.1).
BPCC + equal actual passage-account differences	Ordinal transfer of weak/strict/indifferent protocol rankings without choosing a valuation (Theorem 8.3).
Finite empirical grid + observed weak protocol choices	Sharp counterfactual value bounds and robust out-of-sample protocol predictions (Theorem 9.1).
Finite protocol-price design + quasilinear numeraire	Money-calibrated sharp bounds for passage contrasts (Corollary 9.2).

Table 1

Dependency map for the main representation and its principal implications.

12. Universal carrier and concatenation coherence

Full proof of Proposition 4.1. A nonconstant realizing path gives a walk that uses every ledger arc with its recorded multiplicity. Its nonisolated support is connected and contains the prescribed base point whenever the path is closed. Pairing every arrival with the next departure cancels all internal contributions to outdegree minus indegree; only the initial departure and terminal arrival remain, giving the displayed balance. A zero ledger contains no nonconstant passage, so continuity forces a realizing path to be constant and hence $s = t$.

Conversely, let $L \neq 0$ and replace every positive arc multiplicity by that many parallel labelled arcs. If $s \neq t$, the classical directed Euler-trail criterion applies to the connected nonisolated multigraph with the stated imbalance and gives a trail from s to t . If $s = t$, the support condition places the prescribed base point in the connected Eulerian support, so a closed Euler tour may be started at s . Reading the visited states in order and linearly interpolating adjacent states produces a finite-step path. The arc labels agree with side and direction by construction, including the target endpoint convention. The zero-ledger case is realized by c_s exactly when $s = t$. $\square$

Lemma 12.1 (Concatenation coherence). *Any two parenthesizations of a finite endpoint-compatible concatenation are related by an orientation-preserving piecewise-linear time change. If each factor is independently weakly reparameterized, the resulting concatenations have a common weak-time-change refinement.*

Proof. Give each factor a consecutive subinterval of $[0, 1]$. A parenthesization changes only the lengths assigned to those subintervals, and the unique increasing piecewise-linear map carrying one set of breakpoints to the other identifies the two concatenations. For independently reparameterized factors, refine both breakpoint systems and use the factorwise common monotone refinements on the corresponding cells. Concatenating these maps gives a continuous nondecreasing surjection of $[0, 1]$. $\square$

Proof of Theorem 5.1. Step 1: construction and well-definedness of q . Choose a finite cut of γ containing all turning times and target contacts, and delete constant cells. Each remaining cell is a canonical passage in one module. Define $q(\gamma)$ as the sum of the corresponding generator classes. Any two admissible cuts have a common refinement. Inserting one cut replaces $e_m(a, c)$ by $e_m(a, b) + e_m(b, c)$, so the subdivision relations make the sum independent of the cut. Lemma 12.1 gives invariance under weak time change and additivity under endpoint-compatible concatenation.

Step 2: factorization of an arbitrary admissible valuation. Let $\Phi : \mathcal{P} \rightarrow G$ have the stated properties. On the free generators define

$$\psi_0(e_m(a, b)) = \Phi(p_m(a, b)).$$

A canonical passage through the full interval $(a, c]$ is, up to weak time change, the concatenation of the two canonical subpassages through $(a, b]$ and $(b, c]$. In a toward module the chronological order of these two pieces is reversed, but their values may be added in either order because G is abelian. Hence

$$\psi_0(e_m(a, c)) = \psi_0(e_m(a, b)) + \psi_0(e_m(b, c)).$$

Every subdivision relation therefore lies in $\ker \psi_0$, and ψ_0 descends to a homomorphism $\bar{\Phi} : \mathfrak{L} \rightarrow G$. Decomposing an arbitrary path into canonical cells gives $\Phi = \bar{\Phi} \circ q$.

Step 3: uniqueness. Every generator of $\mathfrak{L}$ is represented by an actual canonical passage. A homomorphism satisfying $\Phi = \bar{\Phi} \circ q$ is therefore fixed on all generators and hence unique.

Step 4: identification with the concrete step ledger. Define

$$e_{T_s}(0, b) \mapsto \mathbf{1}_{[0, b]}, \quad e_{T_s}(a, b) \mapsto \mathbf{1}_{(a, b]}, \quad e_{A_s}(a, b) \mapsto \mathbf{1}_{(a, b]}.$$

Subdivision is respected, so this gives a homomorphism $\Xi : \mathfrak{L} \rightarrow \text{Led}$. It is onto by construction of the four step-function groups. To prove injectivity, collect the finitely many breakpoints occurring in a formal relation and refine to adjacent cells. Evaluation on each punctured cell recovers its coefficient; in a toward module, evaluation at zero separately recovers the target-reaching coefficient. A zero step function therefore has all adjacent-cell coefficients equal to zero, and the subdivision relations then make the original formal element zero. Thus Ξ is an isomorphism and $\Xi \circ q = \ell$. $\square$

13. Restricted path-fiber normal form

Working entirely with actual one-dimensional paths, the proof proceeds through a rooted normal form. Lemma 13.1 converts continuous paths into labelled vertex words. Lemma 13.2 recovers the ordered endpoints of every open ledger. Lemmas 13.3 and 13.4 control where repeated closed excursions may be inserted. Lemma 13.5 then gives the rooted normal form by induction on active edges.

Fix ordered endpoints s, t . Let $\approx_{s,t}$ be the smallest equivalence relation generated by weak time changes and by

$$A * C_1 * C_2 * B \approx_{s,t} A * C_2 * C_1 * B,$$

where C_1, C_2 are closed excursions based at the common joining state. The relation is closed under endpoint-compatible contexts. The global relation $\approx_p$ additionally allows cyclic rotation of a closed path and identifies all constant paths.

Lemma 13.1 (Atomic reduction). *For finitely many finite-step paths there is a common finite state grid containing the target, all endpoints, all turning values, and all module changes. Each path is weak-time-change equivalent to the piecewise-linear realization of a finite vertex word on that path graph. Equality of q is equivalent to equality of every module-labelled directed adjacent-edge multiplicity.*

Proof. Cut each monotone piece whenever it crosses a grid state. After deleting constant subpieces, every cell traverses one adjacent grid edge. Reparameterize each nonconstant cell by normalized distance and assign the cells consecutive time intervals; this produces the canonical piecewise-linear word and a continuous nondecreasing surjection from the original time parameter. The ledger isomorphism in Theorem 5.1 identifies equality of q with equality of the adjacent-cell coefficients. The target-ending toward cell contains the target point, whereas the target-departing away cell excludes it, so the target-attainment multiplicity is preserved. $\square$

Lemma 13.2 (Flow determines open endpoints). *For directed arc multiplicities n_{uv} set*

$$b(v) = \sum_w n_{vw} - \sum_w n_{wv}.$$

Every realizing walk from s to t satisfies

$$b(v) = \mathbf{1}_{\{v=s\}} - \mathbf{1}_{\{v=t\}}.$$

Hence a nonzero balance vector uniquely determines the ordered endpoints. A zero balance vector means that every realization is closed, although its base point may vary.

Proof. Every internal visit contributes once to indegree and once to outdegree. Only the initial departure and terminal arrival remain unpaired. $\square$

Lemma 13.3 (Insertion-point irrelevance). *Let $E : v \rightarrow v$ be a fixed closed excursion and let W visit v . Inserting E^k at any two visits to v gives $\approx_{s,t}$ -equivalent walks.*

Proof. If the two visits enclose a block $C : v \rightarrow v$, the two insertions have the form $A * E^k * C * B$ and $A * C * E^k * B$. Both E^k and C are closed excursions based at v , so one commutation moves the insertion across the block. Repeating this step moves it across any finite number of visits. $\square$

Lemma 13.4 (Rooted lifting). *If $W \approx_{s,t} W'$ and both visit v , then inserting the same fixed excursion $E^k : v \rightarrow v$ at arbitrary v -visits in W and W' preserves rooted equivalence.*

Proof. By Lemma 13.3, move the insertion to a convenient visit before and after each generating step. A weak time change is lifted by using the same expanded atomic word. For a commutation

$$A * C_1 * C_2 * B \longleftrightarrow A * C_2 * C_1 * B,$$

place E^k in an unchanged context if possible. If v occurs only inside C_1 , replace C_1 by the closed excursion obtained after inserting E^k and perform the same commutation; the case of C_2 is symmetric. If both excursions visit v , choose either one. Lemma 13.3 then returns the insertion to the prescribed visits. Induction over the finite operation chain proves the result. $\square$

Lemma 13.5 (Rooted leaf-excursion normal form). *On a finite path graph, fix a connected nonnegative directed-edge ledger and an ordered endpoint pair satisfying Euler balance. Any two Euler trails realizing those data are $\approx_{s,t}$ -equivalent.*

Proof. Induct on the number of active undirected edges. With no active edge, Euler balance forces $s = t$ and the trail is constant. With one active edge, the directed multiplicities and endpoints force a unique alternating vertex word up to weak time change.

Suppose at least two edges are active. The active support is a connected subpath. Choose a leaf u with neighbor v ; in the closed case choose $u \neq s$. Let $a = n_{uv}$, $b = n_{vu}$, and $E = (v \rightarrow u \rightarrow v)$. Euler balance at u gives

$$a - b = \mathbf{1}_{\{u=s\}} - \mathbf{1}_{\{u=t\}}.$$

Because u is a leaf, every occurrence of u other than a global initial or terminal occurrence lies in a contiguous block E . If $u = s \neq t$, there is one unmatched initial edge $u \rightarrow v$ and b copies of E ; if $u = t \neq s$, there is one unmatched terminal edge $v \rightarrow u$ and a copies of E ; otherwise $a = b$ and all leaf uses are copies of E .

Delete the copies of E and the fixed unmatched boundary edge, if any. The resulting trail on the graph with u removed has ledger and endpoints determined solely by the original data: respectively (v, t) , (s, v) , or (s, t) in the three cases. The reduced support remains connected or becomes empty, so induction relates any two reduced trails. Lemma 13.3 concentrates all copies of E at a canonical visit to v , and Lemma 13.4 lifts the equivalence of the reduced trails after inserting the common block E^k . Restoring the identical unmatched boundary edge completes the induction. $\square$

Proof of Theorem 5.2. Necessity. Every generating operation preserves q . Weak time changes and constant-path identification do so by construction. Excursion commutation preserves the sum by additivity and commutativity, and cyclic rotation preserves the labelled edge multiplicities of a closed path. Hence $\gamma \approx_p \eta$ implies $q(\gamma) = q(\eta)$.

Sufficiency for open ledgers. Assume $q(\gamma) = q(\eta)$ and apply Lemma 13.1 on a common finite labelled path graph. If the common ledger has nonzero flow balance, Lemma 13.2 gives the same ordered endpoints for both words. Their directed labelled edge multiplicities and endpoints therefore satisfy the hypotheses of Lemma 13.5, which relates the words by rooted excursion commutations and weak time changes.

Sufficiency for closed and zero ledgers. If every multiplicity is zero, both paths are constant and belong to the globally identified zero-passage class. If the common ledger is nonzero and balanced, both paths are closed. Choose an active vertex r in their common connected support and cyclically rotate both words to the base point r . Lemma 13.5 applies to the rooted closed words; rotating back gives global equivalence.

Target-label preservation. When the leaf removed by Lemma 13.5 is the target, every deleted excursion is $v \rightarrow 0 \rightarrow v$. Its incoming arc is the target-including, target-ending toward cell, while its outgoing arc is the target-excluding, target-departing away cell. Thus it contains exactly one arrival trace and one departure. The induction neither duplicates nor removes target-attainment content. $\square$

14. Canonical envelope and real supporting-valuation separation

Proof of Proposition 6.2. The set $C_{\geq}$ is an additive monoid containing zero. Its symmetric part $\mathcal{N}_{\geq} = C_{\geq} \cap (-C_{\geq})$ is a subgroup. The image of $C_{\geq}$ in the quotient is pointed and additive, hence defines a translation-invariant partial order. Monotonicity, weak-time-change invariance, and admissible additivity of $J_{\geq}$ follow from the definitions and Theorem 5.1.

Assume BPCC and suppose $J_{\geq}(\gamma) \geq J_{\geq}(\eta)$. Then $q(\gamma) - q(\eta) \in C_{\geq} + \mathcal{N}_{\geq} = C_{\geq}$. If $\eta > \gamma$, append that strict reverse comparison to a cone representation of the difference. The aggregate account balances, contradicting BPCC. Completeness therefore gives $\gamma \geq \eta$. Conversely, if BPCC fails, a balanced list with one strict pair puts both directions of that strict difference in the comparison cone, so the quotient collapses the pair and cannot be exact.

Finally, any monotone admissible group-valued valuation sends $C_{\geq}$ to its positive cone and therefore annihilates $\mathcal{N}_{\geq}$. It descends uniquely to a positive homomorphism on $G_{\geq}$. $\square$

Proof of Proposition 6.3. Indifferent differences lie in both C and $-C$, so $I \subseteq C \cap (-C)$. Conversely, suppose $z \in C \cap (-C)$. The two finite conic representations of z use only finitely many passage breakpoints. Refining them to one finite grid identifies all involved account differences with integer vectors in a finite-dimensional rational space. The resulting real conic dependence therefore has a rational certificate in the same face; clearing denominators yields an integer balanced list of actual weak comparisons. BPCC forces every positively weighted comparison in such a two-sided certificate to be indifferent, hence $z \in I$.

If $d(\gamma, \eta) \in C$ and $\eta > \gamma$, the same rationalization and denominator clearing, followed by copies of the strict reverse comparison, gives a balanced list with at least one strict comparison. BPCC rules this out. Thus cone membership of an actual difference is equivalent to the primitive weak comparison. $\square$

Order unit and Archimedeanization. Under strict directional monotonicity, each favorable full-range module passage lies in C . Their sum u contains positive support in every directed module. On a common finite subdivision, any $v \in V = E/I$ is a finite real combination of canonical cells, hence some finite multiple of $e = u + I$ dominates both v and $-v$. Thus e is an order unit. Any two full-support order units dominate one another up to finite positive multiples and therefore generate the same set K^a .

The quotient $\hat{V} = V/N$ by the lineality $N = K^a \cap (-K^a)$, with cone $\hat{K} = K^a/N$, is an Archimedean order-unit space. Its order-unit gauge is a norm and $\hat{K}$ is norm closed. These are the standard Archimedeanization properties; the argument uses no totality of K .

Proof of Theorem 6.5. For an Archimedean order-unit space, normalized positive linear functionals separate the closed positive cone:

$$\hat{v} \in \hat{K} \iff \varphi(\hat{v}) \geq 0 \text{ for every normalized state } \varphi.$$

Indeed, a point outside the closed convex cone is separated by a nonzero continuous linear functional nonnegative on the cone. A nonzero positive functional has strictly positive value at the order unit, because $\varphi(\hat{e}) = 0$ together with $-r\hat{e} \leq \hat{v} \leq r\hat{e}$ for every $\hat{v}$ would force $\varphi = 0$. It can therefore be normalized.

If $\gamma \geq \eta$, Proposition 6.3 puts $d(\gamma, \eta) + I$ in $K \subseteq K^a$, so every normalized supporting valuation is nonnegative on $\hat{d}_{\gamma\eta}$. Conversely, unanimity places the actual difference in K^a , and PVBC returns $\gamma \geq \eta$. Necessity is the reverse implication: if an exact normalized real supporting family represents actual histories and $d(\gamma, \eta) + I \in K^a$, then $n(d + I) + e \in K$ for every n , so every normalized supporting valuation gives $\varphi(d + I) \geq -1/n$ for every n and hence a nonnegative value; exactness yields the weak comparison.

Convexity of the normalized state set is immediate. For each $\hat{v}$, the order-unit property bounds all states on a compact interval. The state set is closed in the corresponding product of compact intervals, so pointwise compactness follows from Tychonoff's theorem. $\square$

15. Primitive balances and restricted-domain geometry

This appendix records two domain facts about BPCC and restricted path geometry. First, fixed menu cardinality does not bound primitive cancellation multiplicity. Second, formal passage differences can fail to compose through a common actual intermediate history.

Fix three states $\ell < 0 < r$ and use coordinates (T_L, A_L, T_R, A_R) . For every integer $m \geq 2$, define

$$\begin{aligned} U_0 &= (1, 0, 0, 0), & U_m &= (m+1, m, 1, 1), \\ V_0 &= (0, 0, 0, 1), & V_m &= (1, 1, m, m+1), \\ P_0 &= (0, 0, 0, 0), & P_1 &= (1, 1, 0, 0). \end{aligned}$$

For columns $x_j \in \mathbb{Z}^4$, let $\tilde{x}_j = (1, x_j)$.

Theorem 15.1 (Fixed-cardinality primitive balances of unbounded degree). *For every $m \geq 2$,*

$$mU_0 + V_m + (m^2 - 1)P_1 = mU_m + V_0 + (m^2 - 1)P_0.$$

In the column order $(U_0, U_m, V_0, V_m, P_0, P_1)$,

$$\ker_{\mathbb{Z}} A_m = \mathbb{Z}z_m, \quad z_m = (m, -m, -1, 1, -(m^2 - 1), m^2 - 1).$$

Hence z_m is a primitive circuit and a Graver element with support six and affine degree $m^2 + m$.

Corollary 15.2 (No uniform bounded-multiplicity substitute for BPCC). *For every integer K , there is a strict total order on a six-history menu that satisfies every BPCC test whose balanced lists contain at most K occurrences on either side, repetitions counted, but violates full BPCC.*

Let $D = \ell(\mathcal{P})$ and $S = D - D$.

Proposition 15.3 (Common-core stabilization of one fixed difference). *If $x - y = g = x' - y'$ with $x, y, x', y' \in D$, then there are nonnegative formal accounts K, K' such that $X = x + K = x' + K'$ and $Y = y + K = y' + K'$ are both reachable.*

Proposition 15.4 (Three-state failure of chain realization). *On the three-state geometry $a < 0 < r$, define*

$$g = \ell(r \rightarrow 0 \rightarrow a) - \ell(a \rightarrow 0 \rightarrow r), \quad h = \ell(0 \rightarrow r) - \ell(r \rightarrow 0).$$

Then $g, h, g + h \in S$, but there are no $x, y, z \in D$ satisfying $x - y = g$ and $y - z = h$.

15.1. Reachability of the six accounts

Write L for the state $\ell < 0$ and R for $r > 0$. The six accounts above are realized by

$$\begin{aligned} U_0 &: L \rightarrow 0, \\ U_m &: L \rightarrow 0, \quad m \text{ copies of } 0 \rightarrow L \rightarrow 0, \quad 0 \rightarrow R \rightarrow 0, \\ V_0 &: 0 \rightarrow R, \\ V_m &: 0 \rightarrow L \rightarrow 0, \quad m \text{ copies of } 0 \rightarrow R \rightarrow 0, \quad 0 \rightarrow R, \\ P_0 &: 0, \\ P_1 &: 0 \rightarrow L \rightarrow 0. \end{aligned}$$

Their coordinates are therefore exactly the displayed (T_L, A_L, T_R, A_R) vectors. The two longest paths use $2m + 3$ atomic edges.

15.2. Kernel, primitivity, and Graver indecomposability

Proof of Theorem 15.1. Let (a, b, c, d, e, f) be an augmented integer relation. We use the column order

$$(U_0, U_m, V_0, V_m, P_0, P_1).$$

The four account rows and the affine row give

$$\begin{aligned} a + (m+1)b + d + f &= 0, \\ mb + d + f &= 0, \\ b + md &= 0, \\ b + c + (m+1)d &= 0, \\ a + b + c + d + e + f &= 0. \end{aligned}$$

Subtracting the second equation from the first gives $a = -b$, and the third gives $b = -md$. Substitution then yields

$$c = -d, \quad e = -(m^2 - 1)d, \quad f = (m^2 - 1)d.$$

Thus every integer kernel vector is

$$d(-m, m, 1, -1, m^2 - 1, -(m^2 - 1)),$$

equivalently an integer multiple of the displayed z_m .

All six entries of z_m are nonzero, so no proper subset of its support carries a nonzero augmented relation. Its coordinates have greatest common divisor one, hence it is a primitive circuit. Because the integer kernel has rank one, every conormal summand of z_m is an integer multiple of z_m . Two nonzero conormal summands adding to z_m would require an integer strictly between zero and one, which is impossible; hence z_m is Graver-indecomposable. Finally, its positive coefficients are m , 1, and $m^2 - 1$, so

$$\text{adeg}(z_m) = m^2 + m, \quad \|z_m\|_\infty = m^2 - 1.$$

Direct substitution gives the displayed affine balance. □

15.3. From the primitive balance to the audit failure

Proof of Corollary 15.2. Choose m with $m^2 + m > K$. Rank the three alternatives in the positive support of z_m , namely U_0, V_m, P_1 , strictly above the three alternatives in its negative support, namely U_m, V_0, P_0 , and order alternatives within each block arbitrarily. Expanding z_m^+ and z_m^- produces a balanced list of length $m^2 + m$ on each side in which every left-side occurrence is strictly preferred to every right-side occurrence. Full BPCC therefore fails.

Now consider any BPCC violation on this six-history menu and cancel alternatives that occur on both aggregate sides. Its reduced augmented coefficient vector belongs to $\ker_{\mathbb{Z}} A_m$, so it equals kz_m for some integer k . Let R be any numerical rank representing the chosen strict order. Every weakly improving comparison contributes a nonnegative rank difference, and at least one contributes a positive difference; hence the aggregate rank difference is positive. For the reduced vector this difference is $k(R \cdot z_m)$.

Because every positive-support alternative is ranked above every negative-support alternative and the two sides of z_m have the same total multiplicity, $R \cdot z_m > 0$. Therefore $k > 0$. The reduced list has at least

$$\|(kz_m)^+\|_1 = k \text{adeg}(z_m) \geq m^2 + m > K$$

occurrences on each side, and an unreduced presentation cannot be shorter. Thus every level- K test passes although full BPCC fails. □

The argument controls expanded list length and coefficient multiplicity. It does not establish an unbounded Fishburn distinct-pair order, an unbounded Bernstein–Onn n -fold Graver complexity for one fixed matrix, computational hardness, or the impossibility of bounds on a separately bounded family of histories.

15.4. Fixed-difference common cores

Proof of Proposition 15.3. Put x, y, x', y' on a common finite grid. On every adjacent grid edge add N arcs in each direction, producing a connected zero-boundary circulation Q_N . For sufficiently large N , both

$$X = x + Q_N, \quad Y = y + Q_N$$

have connected active support and retain the ordered endpoint imbalances of x and y . Proposition 4.1 therefore makes X and Y reachable.

The equality $x - y = x' - y'$ gives $y - y' = x - x'$, and hence

$$Y - y' = Q_N + (y - y') = Q_N + (x - x') = X - x'.$$

For sufficiently large N , this common formal account is coordinatewise nonnegative. Set

$$K = Q_N, \quad K' = Y - y' = X - x'.$$

Then $X = x + K = x' + K'$ and $Y = y + K = y' + K'$, as required. $\square$

15.5. The endpoint obstruction to chain realization

Proof of Proposition 15.4. The displayed expressions show $g, h \in S$, and direct cancellation gives

$$g + h = \ell(0 \rightarrow a) - \ell(a \rightarrow 0) \in S.$$

Apply the endpoint-boundary map

$$\partial\ell(\gamma) = \delta_{\gamma(1)} - \delta_{\gamma(0)}.$$

If $x - y = g$, then

$$\partial x - \partial y = 2(\delta_a - \delta_r).$$

Every path boundary has each coefficient in $\{-1, 0, 1\}$. The coefficient 2 at a therefore forces x to end at a and y to start at a , while the coefficient -2 at r forces x to start at r and y to end at r . Thus y must run from a to r .

If also $y - z = h$, then

$$\partial y - \partial z = 2(\delta_r - \delta_0).$$

The coefficient 2 at r again forces y to end at r , but the coefficient -2 at 0 forces y to start at 0. The same actual path y cannot start both at a and at 0. This contradiction proves the claim. $\square$

16. Punctured closure and regular supporting valuations

Construction of the punctured topology. Let Q° be the collection of all seminorms p on $\widehat{V}$ such that

$$p(\widehat{x}_m(I_n)) \longrightarrow 0$$

for every admissible punctured shrinking sequence $(m, (I_n))$. Let τ° be the locally convex topology generated by Q° . It is the finest locally convex topology with the required geometric convergence property: if another locally convex topology makes every designated passage sequence converge to zero, each of its continuous seminorms belongs to Q° , so that topology is no finer than τ° . Target-reaching sequences belong to the boundary class rather than the designated vanishing class. The topology may be non-Hausdorff; its Hausdorff quotient has the same continuous dual used below.

The continuous dual of the punctured topology. A linear functional $\varphi : \hat{V} \rightarrow \mathbb{R}$ is τ° -continuous if and only if it is punctured regular. Continuity immediately implies $\varphi(\hat{x}_m(I_n)) \rightarrow 0$ whenever the defining passage sequence converges to zero. Conversely, if φ is punctured regular, the seminorm $p_\varphi(v) = |\varphi(v)|$ belongs to Q° and therefore makes φ continuous by construction of τ° . If τ° is non-Hausdorff, the same statement holds on the Hausdorff quotient and the continuous dual is unchanged.

Regular separation in Theorem 7.3. Let $K^\circ = \overline{\hat{K}}^{\tau^\circ}$. If $\gamma \geq \eta$, every positive regular state is nonnegative on $\hat{d}_{\gamma\eta}$. Conversely, suppose $\gamma \not\geq \eta$. Completeness gives $\eta > \gamma$, and PROC implies $\hat{d}_{\gamma\eta} \notin K^\circ$. On the Hausdorff locally convex quotient, separation of the point from the closed convex cone gives a nonzero τ° -continuous functional nonnegative on K° and strictly negative on $\hat{d}_{\gamma\eta}$. Positivity on $\hat{K}$ and the order-unit argument normalize it at $\hat{e}$. The preceding dual characterization makes it a regular state. Hence regular states exactly separate actual histories.

For necessity, if regular states exactly represent actual histories and $\hat{d}_{\gamma\eta} \in K^\circ$, every regular state is nonnegative on that difference because it is continuous and nonnegative on $\hat{K}$. Exactness therefore gives $\gamma \geq \eta$, which is PROC. $\square$

Stieltjes construction in Theorem 7.3. Fix $\varphi \in S_{\text{reg}}$.

Away modules. For an away module define

$$F_m^\varphi(b) = \varphi(\hat{x}_m((0, b])), \quad F_m^\varphi(0) = 0.$$

Subdivision gives finite additivity on half-open intervals. If $b_n \downarrow a$, regularity of $(a, b_n] \downarrow \emptyset$ gives right-continuity. Hence the interval content extends uniquely to a finite Borel Lebesgue–Stieltjes measure. Away target mass is observationally irrelevant and is normalized to zero.

Toward modules. Let $\hat{t}_s(b)$ be the favorable target-reaching generator for $[0, b]$ and set

$$T_s^\varphi(b) = \varphi(\hat{t}_s(b)).$$

For $0 < a < b$, subdivision gives

$$T_s^\varphi(b) - T_s^\varphi(a) = \varphi(\hat{x}_{T_s}((a, b])).$$

Regularity therefore gives right-continuity on $(0, R_s]$. Monotonicity yields the finite limit

$$\alpha_s^\varphi = \lim_{b \downarrow 0} T_s^\varphi(b) \geq 0.$$

Subtracting this residue produces a right-continuous punctured cumulative function. Its Stieltjes measure augmented by $\alpha_s^\varphi \delta_0$ represents both punctured toward passages and target-reaching passages. Target-reaching sequences were excluded from the definition of τ° , so nothing forces α_s^φ to vanish.

Every finite-step module profile is a finite nonnegative integer combination of canonical interval indicators. Additivity therefore extends the four measure formulas to every path. Canonical interval and target-reaching values determine the toward measures and the punctured away measures uniquely; the zero-away-atom convention completes the normalization. $\square$

Proof of Corollary 7.4. Canonical punctured and target-reaching generators determine every regular state, so the state-to-measure map is injective. If the normalized regular state family is a singleton $\{\varphi^*\}$, then $J_{\text{reg}} = \ker(\varphi^* \circ \rho)$ has codimension one. Conversely, if E/J_{reg} is one-dimensional, every regular state factors through that quotient and normalization at the common order unit fixes the remaining positive scalar. This proves equivalence of the state, measure, and codimension statements. $\square$

Lemma 16.1 (Target-interface compatibility). *For side s and any finite-step path,*

$$A_s(\gamma) = h_\gamma^{T_s}(0) = \lim_{r \downarrow 0} N_s^T(r; \gamma).$$

Under subdivision, a target-reaching toward passage has trace $1 = 1 + 0$: only the subpassage ending at the target retains the attainment. Consequently, under every regular supporting valuation Λ , the target contribution to $U_\Lambda(\gamma)$ is exactly $\alpha_s^\Lambda A_s(\gamma)$, and a target-touching loop of radius ϵ has value converging to α_s^Λ .

Proof. Every arrival from side s contributes one target-ending generator $\mathbf{1}_{[0,b]}$, while every punctured toward generator vanishes at zero. A finite-step path has a smallest positive breakpoint on each side whenever it has a nonzero near-target trace. Its toward crossing profile is therefore constant on a sufficiently small punctured neighborhood of zero, and that constant equals the number of attainments from side s .

If $0 < a < b$, subdivision writes $[0, b]$ as the accounting sum $[0, a] \sqcup (a, b]$. The first generator has target trace one and the second has trace zero, so subdivision preserves one attainment rather than duplicating it. Under a regular measure system, a target-touching loop additionally contains one toward and one away punctured passage through $(0, \varepsilon]$; both punctured masses converge to zero, leaving the target atom. $\square$

17. Potential, gross-passage, and attainment identities

Proof of Lemma 5.3. For fixed side s and level $r > 0$, consider the binary state $\chi_s(\gamma(t), r)$. Each toward crossing changes it from one to zero, and each away crossing changes it from zero to one. Telescoping over the finite crossings gives

$$\chi_s(\gamma(1), r) - \chi_s(\gamma(0), r) = N_s^A(r; \gamma) - N_s^T(r; \gamma),$$

which is the stated identity. $\square$

Proof of Theorem 8.1. For any Borel set E on the punctured side,

$$|\kappa_s(E)| \leq (\lambda_{T_s}^\circ(E) + \lambda_{A_s}^\circ(E))/2 = \mu_s(E).$$

Hence $\kappa_s \ll \mu_s$ and its Radon–Nikodym derivative satisfies $|\theta_s| \leq 1$ μ_s -almost everywhere.

Algebraically,

$$\int N_s^T d\lambda_{T_s}^\circ - \int N_s^A d\lambda_{A_s}^\circ = \int D_s^\gamma d\mu_s + \int V_s^\gamma d\kappa_s.$$

By Lemma 5.3, summing the first integrals over the two sides gives

$$\sum_s [\mu_s((0, |\gamma(0)|]) \mathbf{1}_{\{\gamma(0) \text{ on } s\}} - \mu_s((0, |\gamma(1)|]) \mathbf{1}_{\{\gamma(1) \text{ on } s\}}] = P(\gamma(1)) - P(\gamma(0)).$$

Lemma 16.1 supplies the atom term. The inverse transformation is

$$\lambda_{T_s}^\circ = \mu_s + \kappa_s, \quad \lambda_{A_s}^\circ = \mu_s - \kappa_s, \quad \lambda_s^T(\{0\}) = \alpha_s,$$

so, after the zero-away-atom normalization and $P(0) = 0$, the induced decomposition parameters are unique. $\square$

For a canonical off-target round trip, $D_s = 0$ and $V_s = 2\mathbf{1}_{(a,b]}$, giving $U(C_{s;a,b}) = 2\kappa_s((a, b])$. A target-return loop of radius ε has $D_s = 0$, $V_s = 2\mathbf{1}_{(0,\varepsilon]}$, and one arrival, so

$$U(C_{\varepsilon,s}) = 2\kappa_s((0, \varepsilon]) + \alpha_s.$$

Finite signed-measure continuity gives $\kappa_s((0, \varepsilon]) \rightarrow 0$, proving Corollary 8.4. All off-target round trips vanish exactly when $\kappa_s = 0$ on both sides; target-return loops then vanish exactly when $\alpha_s = 0$, which is the loop characterization used in Corollary 8.5.

CRedit authorship contribution statement

Yushang Cheng: Conceptualization, Methodology, Formal analysis, Writing – original draft, and Writing – review and editing.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data were used for the research described in this article.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT to assist with language editing, structural checking, and LaTeX formatting. The author reviewed and edited the resulting material and takes full responsibility for the content of the article.

References

- Abraham, J., Kotzig, A., 1980. Transformations of euler tours, in: *Combinatorics 79, Part I*. North-Holland. volume 8 of *Annals of Discrete Mathematics*, pp. 65–69.
- Becker, G.S., Grossman, M., Murphy, K.M., 1994. An empirical analysis of cigarette addiction. *American Economic Review* 84, 396–418. URL: <https://www.jstor.org/stable/2118059>.
- Becker, G.S., Murphy, K.M., 1988. A theory of rational addiction. *Journal of Political Economy* 96, 675–700. doi:10.1086/261558.
- Bernheim, B.D., Rangel, A., 2004. Addiction and cue-triggered decision processes. *American Economic Review* 94, 1558–1590. doi:10.1257/0002828043052222.
- Berstein, Y., Onn, S., 2009. The graver complexity of integer programming. *Annals of Combinatorics* 13, 289–296. doi:10.1007/s00026-009-0029-6.
- Candogan, O., Menache, I., Ozdaglar, A., Parrilo, P.A., 2011. Flows and decompositions of games: Harmonic and potential games. *Mathematics of Operations Research* 36, 474–503. doi:10.1287/moor.1110.0500.
- Chateauneuf, A., 1985. On the existence of a probability measure compatible with a total preorder on a boolean algebra. *Journal of Mathematical Economics* 14, 43–52. doi:10.1016/0304-4068(85)90025-4.
- Chateauneuf, A., Jaffray, J.Y., 1984. Archimedean qualitative probabilities. *Journal of Mathematical Psychology* 28, 191–204. doi:10.1016/0022-2496(84)90026-9.
- Chateauneuf, A., Maccheroni, F., Marinacci, M., Tallon, J.M., 2005. Monotone continuous multiple priors. *Economic Theory* 26, 973–982. doi:10.1007/s00199-004-0540-2.
- Chateauneuf, A., Wakker, P.P., 1993. From local to global additive representation. *Journal of Mathematical Economics* 22, 523–545. doi:10.1016/0304-4068(93)90002-3.
- Chen, K.T., 1954. Iterated integrals and exponential homomorphisms. *Proceedings of the London Mathematical Society* s3-4, 502–512. doi:10.1112/plms/s3-4.1.502.
- Dixit, A.K., 1989. Entry and exit decisions under uncertainty. *Journal of Political Economy* 97, 620–638. doi:10.1086/261619.
- Dubra, J., Maccheroni, F., Ok, E.A., 2004. Expected utility theory without the completeness axiom. *Journal of Economic Theory* 115, 118–133. doi:10.1016/S0022-0531(03)00166-2.
- Evren, O., Ok, E.A., 2011. On the multi-utility representation of preference relations. *Journal of Mathematical Economics* 47, 554–563. doi:10.1016/j.jmateco.2011.07.003.
- Fishburn, P.C., 2001. Cancellation conditions for finite two-dimensional additive measurement. *Journal of Mathematical Psychology* 45, 2–26. doi:10.1006/jmps.1999.1285.
- Fleischner, H., Sabidussi, G., Wenger, E., 1992. Transforming eulerian trails. *Discrete Mathematics* 109, 103–116. doi:10.1016/0012-365X(92)90281-J.
- Frey, B.S., Benz, M., Stutzer, A., 2004. Introducing procedural utility: Not only what, but also how matters. *Journal of Institutional and Theoretical Economics* 160, 377–401. doi:10.1628/0932456041960560.
- Frey, B.S., Stutzer, A., 2005. Beyond outcomes: Measuring procedural utility. *Oxford Economic Papers* 57, 90–111. doi:10.1093/oep/gpi002.
- Giné, X., Karlan, D., Zinman, J., 2010. Put your money where your butt is: A commitment contract for smoking cessation. *American Economic Journal: Applied Economics* 2, 213–235. doi:10.1257/app.2.4.213.
- Graver, J.E., 1975. On the foundations of linear and integer linear programming i. *Mathematical Programming* 9, 207–226. doi:10.1007/BF01681344.
- Gruber, J., Köszegi, B., 2001. Is addiction rational? theory and evidence. *Quarterly Journal of Economics* 116, 1261–1303. doi:10.1162/003355301753265570.
- Hamblly, B., Lyons, T., 2010. Uniqueness for the signature of a path of bounded variation and the reduced path group. *Annals of Mathematics* 171, 109–167. doi:10.4007/annals.2010.171.109.
- Hsiaw, A., 2013. Goal-setting and self-control. *Journal of Economic Theory* 148, 601–626. doi:10.1016/j.jet.2012.08.001.
- Jiang, X., Lim, L.H., Yao, Y., Ye, Y., 2011. Statistical ranking and combinatorial hodge theory. *Mathematical Programming* 127, 203–244. doi:10.1007/s10107-010-0419-x.

- Koch, A.K., Nafziger, J., 2011. Self-regulation through goal setting. *Scandinavian Journal of Economics* 113, 212–227. doi:10.1111/j.1467-9442.2010.01641.x.
- Kopylov, I., 2007. Subjective probabilities on “small” domains. *Journal of Economic Theory* 133, 236–265. doi:10.1016/j.jet.2005.12.004.
- Kőszegi, B., Rabin, M., 2006. A model of reference-dependent preferences. *The Quarterly Journal of Economics* 121, 1133–1165. doi:10.1093/qje/121.4.1133.
- Kraft, C.H., Pratt, J.W., Seidenberg, A., 1959. Intuitive probability on finite sets. *The Annals of Mathematical Statistics* 30, 408–419. doi:10.1214/aoms/1177706260.
- Krantz, D.H., Luce, R.D., Suppes, P., Tversky, A., 1971. *Foundations of Measurement, Volume I: Additive and Polynomial Representations*. Academic Press, New York.
- Łochowski, R.M., 2017. On a generalisation of the banach indicatrix theorem. *Colloquium Mathematicum* 147, 301–318. doi:10.4064/cm6583-3-2017.
- Loewenstein, G.F., Prelec, D., 1993. Preferences for sequences of outcomes. *Psychological Review* 100, 91–108. doi:10.1037/0033-295X.100.1.91.
- Luce, R.D., 1967. Sufficient conditions for the existence of a finitely additive probability measure. *The Annals of Mathematical Statistics* 38, 780–786. doi:10.1214/aoms/1177698871.
- Mackenzie, A., 2021. On atom-swarmling and luce’s theorem for probabilistic beliefs. *Economic Theory Bulletin* 9, 67–74. doi:10.1007/s40505-020-00194-5.
- Mielke, A., Roubíček, T., 2015. Rate-Independent Systems: Theory and Application. volume 193 of *Applied Mathematical Sciences*. Springer, New York. doi:10.1007/978-1-4939-2706-7.
- Minardi, S., Savochkin, A., 2024. Time for memorable consumption. *Games and Economic Behavior* 148, 296–322. doi:10.1016/j.geb.2024.09.010.
- Ok, E.A., 2002. Utility representation of an incomplete preference relation. *Journal of Economic Theory* 104, 429–449. doi:10.1006/jeth.2001.2814.
- Orphanides, A., Zervos, D., 1998. Myopia and addictive behaviour. *Economic Journal* 108, 75–91. doi:10.1111/1468-0297.00274.
- Parikh, R.J., 1966. On context-free languages. *Journal of the ACM* 13, 570–581. doi:10.1145/321356.321364.
- Qin, W.Z., Rommeswinkel, H., 2022. Additive representations on a simplex. *Journal of Mathematical Economics* 103, 102769. doi:10.1016/j.jmateco.2022.102769.
- Scott, D., 1964. Measurement structures and linear inequalities. *Journal of Mathematical Psychology* 1, 233–247. doi:10.1016/0022-2496(64)90002-1.
- Suranovic, S.M., Goldfarb, R.S., Leonard, T.C., 1999. An economic theory of cigarette addiction. *Journal of Health Economics* 18, 1–29. doi:10.1016/S0167-6296(98)00037-X.
- Tatakis, C., Thoma, A., 2015. Graver degrees are not polynomially bounded by true circuit degrees. *Journal of Pure and Applied Algebra* 219, 2658–2665. doi:10.1016/j.jpaa.2014.09.021.
- Tversky, A., Kahneman, D., 1991. Loss aversion in riskless choice: A reference-dependent model. *The Quarterly Journal of Economics* 106, 1039–1061. doi:10.2307/2937956.
- Varey, C., Kahneman, D., 1992. Experiences extended across time: Evaluation of moments and episodes. *Journal of Behavioral Decision Making* 5, 169–185. doi:10.1002/bdm.3960050303.
- Villegas, C., 1964. On qualitative probability σ -algebras. *The Annals of Mathematical Statistics* 35, 1787–1796. doi:10.1214/aoms/1177700400.
- Wakker, P.P., 1989. *Additive Representations of Preferences: A New Foundation of Decision Analysis*. Kluwer Academic Publishers, Dordrecht. doi:10.1007/978-94-015-7815-8.
- Winston, G.C., 1980. Addiction and backsliding: A theory of compulsive consumption. *Journal of Economic Behavior and Organization* 1, 295–324. doi:10.1016/0167-2681(80)90009-8.